

The original primitive source band and residual boundary return

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1 The original primitive source and its adjacent Gamma metric

This calculation continues the complete LCA15–21 polynomial quotient construction. Its new conclusions are the explicit codimension-three truncation, its exact rank-one native metric change, and a finite coefficient-product bound for that change. All spaces and measures below are the original ones. The connecting target metric and its later long return remain the literal target metrics; the source computation here is at the two specified low cutoffs.

1.1 Original objects and the polynomial quotient map

Fix the actual simple quartet, its invertible period matrix H , logarithm branch, and full conductor coefficients. Retain

$$\begin{aligned} \beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, & \delta, \gamma &> 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, & \mu_r &= E_A^{(r)}(0), & v = \text{ord}_0 E_A \leq 80, & \mu_v \neq 0, \\ L_i(Z, W) &= (H(ZW, Z, W, 1)^T)_i, & b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b} z}, \\ b_i &= B_i(0) = \sum_{a,b=0}^4 b_{i,a,b}, & \sum_{i=1}^4 D_i(z) B_i(z) &= 1. \end{aligned} \tag{PDS1}$$

The D_i are the proved original entire Bezout expressions. In particular at least one b_i is nonzero. Keep the full biform conductor, including repeated components and zero edge coefficients. Set

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_k &= (k + 1)^2, & Q &= (k + 5)^2, \\ e &\in \{0, 1\}, & D_e &= Q - v - 1 + e, & \ell_e &= 8k + 23 + e, & d_e &= \ell_e + 1, \\ \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & (\mathcal{T}_{AP})(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ (\mathcal{T}_{B_i} p)(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}). \end{aligned} \tag{PDS2}$$

All dual functionals are complex-linear. For a finite germ F through degree D_e , let $\Lambda_F(S'^r) = F^{(r)}(0)$. Let $\mathcal{H}_e = \mathbb{C}[S']_{\leq D_e}^\vee$, and let $\mathcal{S}_e \subset \mathcal{H}_e$ be the jets of the original analytic germs

$$F(z) = \frac{N(z)}{E_A(z)}, \quad N(z) = \sum_{a,b=0}^k n_{ab} e^{\beta_{k,a,b} z}, \quad N^{(r)}(0) = 0 \quad (0 \leq r < v). \tag{PDS3}$$

Define the unchanged relation polynomials and quotient coordinates by

$$r_a(S') = \mathcal{T}_A(\chi_k(S) S^a), \quad 0 \leq a \leq \ell_e, \quad \pi_e(F) = (\Lambda_F(r_a))_{a=0}^{\ell_e}. \tag{PDS4}$$

Then π_e is onto and $\ker \pi_e = \mathcal{S}_e$. Here is the complete kernel argument. The distinct upper k nodes give a Vandermonde matrix, whose first v rows have rank v since $v \leq 80 < q_k$. The numerator space in PDS3 therefore has dimension $q_k - v$. Its jet map is injective: if F vanishes through degree D_e , then $N = E_A F$ vanishes through degree $D_e + v \geq q_k - 1$; the first q_k Vandermonde rows force every $n_{ab} = 0$. The Leibniz identity gives $\Lambda_F(\mathcal{T}_A p) = \Lambda_{E_A F}(p)$, so all r_a annihilate PDS3. Finally

$$\mathcal{T}_A S^m = \sum_{r=0}^m \binom{m}{r} \mu_r S'^{m-r}, \quad \deg r_a = q_k + a - v, \quad \text{lc}(r_a) = \mu_v \binom{q_k + a}{v}. \quad (\text{PDS5})$$

These successive degrees prove independence. Their common kernel has dimension $D_e + 1 - d_e = q_k - v$, proving the assertion. This is precisely the LCA16 quotient, with no new row norm. Every original lower degree- $(k - 8)$ exponential is in $\mathcal{S}_e$, since its product with E_A has an allowed degree- k numerator. Thus taking the original lower-gauge quotient first and then quotienting by its analytic-source image gives exactly the same affine fibres as $\mathcal{H}_e/\mathcal{S}_e$.

1.2 The full original Gamma dual metric

The source member of the two original order pairs is $s = 1$ or $s = k$, and its centre is $c = k/2 - 4$. Keep its entire measure and recurrence:

$$M_s = (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = M_s n!(s/2)_n, \\ p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1+s/2)p_{n-1,s}, \quad \phi_n(S') = p_{n,s}((S' - c)/i)/\sqrt{h_{n,s}}. \quad (\text{PDS6})$$

These are the original orthonormal polynomials for the norm $\|p\|_{s,c}^2 = \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y)$, as established in the original Gamma construction reproduced in the cumulative providers. The coefficient isometry uses this complete measure, rather than dividing its mass out. Let B_e be the triangular matrix defined by $S'^r = \sum_{n=0}^{D_e} (B_e)_{nr} \phi_n(S')$ and let

$$C_e = (B_e \text{coeff}_{S'} r_a)_{a=0}^{\ell_e}, \quad H_e = C_e^* C_e > 0, \quad G_e = \overline{H_e^{-1}}. \quad (\text{PDS7})$$

The columns of C_e are polynomial coefficients in the original orthonormal frame. For $w = (\Lambda_F(\phi_n))_n$ the literal quotient row map is $x = C_e^\top w$. Its minimum-norm lift is $w = \overline{C_e} \overline{H_e}^{-1} x$ and has squared norm $x^* G_e x$. Indeed $C_e^\top \overline{C_e} = \overline{H_e}$, and the displayed lift is orthogonal to $\ker C_e^\top$. This proves both the full complex-linear dual orientation and the quotient metric, including every conjugation in PDS7. The same metric results from the two successive minima over the original lower gauges and analytic source, by the equality of fibres proved after PDS5.

The degree-one primitive condition at the first endpoint is empty. At the second endpoint its four original rows are

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = (\text{coeff}_S R_i)_{i=1}^4 \in \mathbb{C}^{d_1 \times 4}, \quad \mathcal{P}_0 = \mathbb{C}^{d_0}, \quad \mathcal{P}_1 = \ker W^\top \subset \mathbb{C}^{d_1}. \quad (\text{PDS8})$$

Here $\mathcal{P}_e$ denotes the primitive quotient source, not a biform section space. All degree- k roots become degree- j roots after every degree-four shift; hence the numerator is divisible by χ_k and $\deg R_i \leq \ell_1$. Translation sums commute, so $\mathcal{T}_A(\chi_k R_i) = \mathcal{T}_{B_i} \mathcal{T}_A \chi_j$. This is exactly the original positive degree-zero exterior differential, and its row is $W^\top$, as in LCA18–19. The full independence proof LCA20–21 applies to the actual invertible period, without a genericity exclusion: the four L_i^4 are independent on the rank-four Segre quadric, and the 25 distinct translations of the degree- $Q - v \geq 116$ polynomial $\mathcal{T}_A \chi_j$ give an injective Vandermonde moment map. Thus $\text{rank } W = 4$. The complete LCA20–21 proof is retained as a provider in the cumulative source, rather than replacing this fact by a new period assumption.

The metric on $\mathcal{P}_e$ is the restriction of G_e . At $e = 1$ its exact polynomial dual is

$$\text{span}(r_0, \dots, r_{\ell_1}) / \text{span}(\mathcal{T}_A(\chi_k R_i))_{i=1}^4.$$

To verify the metric as well as the vector-space statement, choose any fixed full basis matrix Z_1 of $\ker W^\top$. A coefficient vector f of a relation polynomial restricts to the functional $Z_1^\top f$ on this basis. Minimizing $f^* H_1 f$ over all lifts with $Z_1^\top f = a$ gives

$$\min_{Z_1^\top f = a} f^* H_1 f = a^* (Z_1^\top H_1^{-1} Z_1)^{-1} a = a^* \overline{(Z_1^* G_1 Z_1)^{-1}} a. \quad (\text{PDS9})$$

The minimizer follows by the adjoint of $Z_1^\top H_1^{-1/2}$; subtracting it leaves an orthogonal kernel vector. The kernel of $Z_1^\top$ is exactly $\text{im } W$, by dimensions and $W^\top Z_1 = 0$. This proves the stated original quotient metric and its exact dual morphism, with no deletion of the four row polynomials.

1.3 The exact codimension-three truncation

The coefficient of S^{ℓ_1} in R_i is b_i , since both χ_j and χ_k are monic. Choose a fixed original index i_* with $b_{i_*} \neq 0$, write $b_* = b_{i_*}$, and retain

$$R_{i_*}(S) = \sum_{a=0}^{\ell_0} (u_*)_a S^a + b_* S^{\ell_1}, \quad C_i(S) = R_i(S) - \frac{b_i}{b_*} R_{i_*}(S) \quad (i \neq i_*). \quad (\text{PDS10})$$

These displayed combinations are exact coordinates for the intersection of the original four-row span with lower degree; the original rows and coefficients in PDS8 remain fixed. The three C_i have degree at most ℓ_0 and are independent: a relation among them would give a relation among all four R_i , whose coefficients on indices other than i_* are unchanged. Let V have the three coefficient columns C_i , in the original index order with i_* omitted. Then

$$\mathcal{Z} = \ker V^\top \subset \mathbb{C}^{d_0}, \quad \dim \mathcal{Z} = d_0 - 3 = 8k + 21, \quad E = \begin{bmatrix} I_{d_0} \\ -u_*^\top/b_* \end{bmatrix}, \quad (\text{PDS11})$$

$$\text{pr}_0 : \mathcal{P}_1 \xrightarrow{\simeq} \mathcal{Z}, \quad \text{pr}_0^{-1} = E|_{\mathcal{Z}}.$$

In fact the i_* row of $W^\top(x_0, x_{\text{top}}) = 0$ gives $x_{\text{top}} = -u_*^\top x_0/b_*$. Substitution in the other three rows gives precisely $V^\top x_0 = 0$. The first d_0 relation polynomials in PDS4 are unchanged at the two cutoffs, so pr_0 is also the actual jet truncation on the primitive quotient source. If an upper analytic source is added its truncated jet is an old analytic source; hence this map is well-defined on the original quotient. Surjectivity and the displayed inverse follow from the equations just proved.

1.4 One rank-one update, with every frame retained

Choose an arbitrary fixed full column basis Z of $\mathcal{Z}$; use EZ at the upper endpoint. For the same source order s and centre c , the old orthonormal polynomials are unchanged when the cutoff increases. In particular the old C_0 is padded by a zero last row in C_1 . Write the exact block matrix and Schur scalar

$$H_1 = \begin{bmatrix} H_0 & b \\ b^* & a \end{bmatrix}, \quad b = C_0^* c_{\text{top}}, \quad a = c_{\text{top}}^* c_{\text{top}}, \quad \sigma = a - b^* H_0^{-1} b > 0, \quad (\text{PDS12})$$

where c_{top} is the full coefficient column of r_{ℓ_1} . Padded columns are understood in the products. Define

$$\eta = -u_*^\top/b_* - b^\top \overline{H_0}^{-1}, \quad K_0 = Z^* \overline{H_0}^{-1} Z, \quad K_1 = (EZ)^* \overline{H_1}^{-1} (EZ). \quad (\text{PDS13})$$

Then the complete metric comparison is

$$\boxed{K_1 = K_0 + \frac{(\eta Z)^* (\eta Z)}{\sigma}, \quad \frac{\det K_1}{\det K_0} = 1 + \frac{(\eta Z) K_0^{-1} (\eta Z)^*}{\sigma} \geq 1.} \quad (\text{PDS14})$$

For proof, block inversion of $\overline{H_1}$ gives for every (x_0, x_{top})

$$\begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix}^* \overline{H_1}^{-1} \begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix} = x_0^* \overline{H_0}^{-1} x_0 + \frac{|x_{\text{top}} - b^\top \overline{H_0}^{-1} x_0|^2}{\sigma}. \quad (\text{PDS15})$$

Insert PDS11 and then $x_0 = Zz$. This proves the matrix equality. The determinant identity follows by taking the determinant of $I + K_0^{-1/2}(\eta Z)^*(\eta Z)K_0^{-1/2}/\sigma$; this rank-at-most-one operator has its only possible nonzero eigenvalue $(\eta Z)K_0^{-1}(\eta Z)^*/\sigma$. The frame Z has not been changed: replacing it by ZT changes both determinants by the same $|\det T|^2$. PDS14 also proves that the actual truncation $\mathcal{P}_1 \rightarrow \mathcal{Z}$ is a contraction for the displayed native metrics, and locates its entire metric defect in this one explicit row.

1.5 An explicit coefficient-product upper estimate

Set $D = D_1 = Q - v$ and keep the literal original polynomial

$$r_*(S') = \mathcal{T}_A(\chi_k R_{i_*})(S') = \mathcal{T}_{B_{i_*}} \mathcal{T}_A \chi_j(S'), \quad \lambda_* = \text{lc } r_* = b_* \mu_v \binom{Q}{v} \neq 0. \quad (\text{PDS16})$$

There is an exact finite upper bound

$$1 \leq \frac{\det K_1}{\det K_0} \leq \frac{\|r_*\|_{s,c}^2}{|b_*|^2 \sigma} \leq \frac{\|r_*\|_{s,c}^2}{|\lambda_*|^2 h_{D,s}}. \quad (\text{PDS17})$$

To prove the first upper inequality, the orthogonal projector onto $\text{im}(\overline{H}_0^{-1/2} Z)$ gives $ZK_0^{-1}Z^* \leq \overline{H}_0$. Thus the numerator in PDS14 is at most $\eta \overline{H}_0 \eta^*$. Let $v_* = u_*/b_* + H_0^{-1}b$. Then $\eta = -v_*^\top$, so this upper bound is $v_*^* H_0 v_*$. The orthogonal projection of r_* onto the old relation span has coefficient vector $b_* v_*$, whereas its orthogonal remainder is b_* times the remainder of r_{ℓ_1} . Consequently

$$\|r_*\|_{s,c}^2 = |b_*|^2 (v_*^* H_0 v_* + \sigma). \quad (\text{PDS18})$$

This proves the first inequality of PDS17. Every old relation has degree at most $D - 1$. The coefficient of ϕ_D in r_{ℓ_1} is $i^D \mu_v \binom{Q}{v} \sqrt{h_{D,s}}$; this uses its actual leading coefficient and the phase $i^{-D}/\sqrt{h_{D,s}}$ in PDS6. This coefficient remains in its orthogonal remainder. Hence $\sigma \geq |\mu_v \binom{Q}{v}|^2 h_{D,s}$, proving the second inequality of PDS17 with its original scalar intact.

For a completely finite coefficient bound define

$$L = \max\{1, |c| + 2(D + 1 + s/2)\}, \quad \Theta_{a,b,r,t;u,w} = \beta_{8,a,b} + \beta_{4,r,t} - \beta_{j,u,w},$$

$$P_* = \sum_{a,b=0}^8 \sum_{r,t=0}^4 |a_{ab} b_{i_*,r,t}| \prod_{u,w=0}^j (L + |\Theta_{a,b,r,t;u,w}|), \quad U_* = \frac{M_s P_*^2}{|b_* \mu_v \binom{Q}{v}|^2 h_{D,s}}. \quad (\text{PDS19})$$

The L in this subsection is a numerical bound and is not the jet length denoted L in LCA2. Then

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_*. \quad (\text{PDS20})$$

Here is a direct proof that PDS19 bounds the original Gamma norm. In the orthonormal polynomial basis in y , multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$, by PDS6 and $h_{n+1,s}/h_{n,s} = (n+1)(n+s/2)$. On polynomials of degree at most $D-1$ both weighted shifts have norm at most $D+1+s/2$; their sum has norm at most $2(D+1+s/2)$. Thus multiplication by $S' = c + iy$, through degree D , has norm at most L . Starting with the constant polynomial, whose squared norm is M_s , proves

$$\|p\|_{s,c} \leq \sqrt{M_s} \sum_{m=0}^D |[S'^m]p| L^m \quad (\deg p \leq D). \quad (\text{PDS21})$$

The exact expansion of PDS16 is

$$r_*(S') = \sum_{a,b=0}^8 \sum_{r,t=0}^4 a_{ab} b_{i_*,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w}). \quad (\text{PDS22})$$

The weighted coefficient ℓ^1 norm is submultiplicative: for two polynomials its coefficient convolution and the triangle inequality give $\sum_m |(fg)_m| L^m \leq (\sum_m |f_m| L^m)(\sum_m |g_m| L^m)$. Applying this identity to every factor

in PDS22 bounds the weighted coefficient sum of r_* by P_* . Its coefficients above degree $D = Q - v$ vanish by the exact moment identities in PDS5; bounding the full degree- Q expansion before those cancellations is still an upper bound for the unchanged degree- D polynomial. PDS21 now gives $\|r_*\|_{s,c}^2 \leq M_s P_*^2$, and PDS17 proves PDS20. In particular $U_* \geq 1$ is a consequence, rather than an extra assumption on the original coefficients.

There is also an explicit uniform growth bound in either original source-order family. Define the fixed positive numbers

$$\begin{aligned} R_* &= \sqrt{(1/2 + \delta)^2 + \gamma^2}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \quad B_{\text{abs}} = \sum_{r,t=0}^4 |b_{i_*,r,t}|, \\ C_* &= 6 + 17R_*, \quad C_{\text{abs}} = \max \left\{ 1, \frac{2(A_{\text{abs}}B_{\text{abs}})^2}{|b_*\mu_v|^2} \right\}. \end{aligned} \tag{PDS23}$$

The original nodes satisfy $|\operatorname{Re} \beta_{n,a,b}| \leq n(1/2 + \delta)$ and $|\operatorname{Im} \beta_{n,a,b}| \leq n\gamma$. Both bounds are attained together at $(a, b) = (n, 0)$ and (n, n) ; thus $\max_{a,b} |\beta_{n,a,b}| = nR_*$, and therefore $|\Theta| \leq (k + 16)R_*$. For $s = 1$ or $s = k$, $D \leq Q$, $k \leq Q$, and $|c| \leq k/2 + 4$, so $L \leq 2Q + 3k/2 + 6 \leq 6(Q + 1)$. Also $k + 16 \leq 17(Q + 1)$. Every factor in PDS19 is consequently at most $C_*(Q + 1)$, giving $P_* \leq A_{\text{abs}}B_{\text{abs}}[C_*(Q + 1)]^Q$. The full factorial denominator gives a sharper bound than dropping its growth. For $s \geq 1$,

$$D!(s/2)_D \geq D!(1/2)_D = \frac{(2D)!}{4^D} \geq \left(\frac{D}{\exp(1)} \right)^{2D}.$$

The product identity follows by separating the even and odd factors of $(2D)!$. The last inequality follows from $\sum_{r=1}^{2D} \log r \geq \int_1^{2D} \log x dx = 2D \log(2D) - 2D + 1$. Since $D = Q - v$, $Q \geq 196$ and $v \leq 80$ imply $Q + 1 \leq 2D$, the exact PDS19 bound therefore satisfies

$$\begin{aligned} U_* &\leq \frac{(A_{\text{abs}}B_{\text{abs}})^2}{|b_*\mu_v|^2} C_*^{2Q} (Q + 1)^{2Q} \left(\frac{\exp(1)}{D} \right)^{2D} \\ &\leq C_{\text{abs}} (Q + 1)^{2v} (2 \exp(1) C_*)^{2Q}. \end{aligned}$$

Here $\binom{Q}{v} \geq 1$ was used only in an upper bound; the full binomial, mass and factorial denominator remain in PDS19. For the second inequality split $(Q + 1)^{2Q}$ into its $2v$ and $2D$ powers and use $Q + 1 \leq 2D$, $D \leq Q$ and $C_* \geq 6$. Consequently the proved finite bound is

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_* \leq \log C_{\text{abs}} + 2v \log(Q + 1) + 2Q \log(2 \exp(1) C_*) = O_{\text{actual}}(Q) = o(kq_k).$$

(PDS24)

The last limit uses $Q/q_k \rightarrow 1$, fixed v , and $1/k \rightarrow 0$ for $k \equiv 1 \pmod{4}$. The constants are fixed by the actual quartet and period, independent of k and of the choice between the two original source-order families. The sharp finite PDS19 retains the full mass, phase-dependent nodes, factorials, conductor coefficients and leading scalar; PDS24 proves the scale of its upper bound without assuming an inverse estimate.

The maps PDS11 and PDS14 now give the primitive part of the low-cutoff source comparison inside the original LCA connecting sequence. The full cohomology additionally has the retained $\ker U_e$ constituent from LCC. Passing these source metrics through the actual connecting map requires the original target quotient Gram; PDS9 specifies the exact source dual to be used in that calculation, and PDS20–24 control its adjacent source correction. None of these equalities identifies a later target coefficient flag with a common compatible source at all cutoffs.

2 The original connecting image and its full native target determinant

Keep the complete CNC, LCC and LCA data, including the nonreduced A^{hom} , every original coefficient and the exact $E_A(z) = \sum a_{ab} e^{\beta_{s,a,b} z}$. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_a &= (a + 1)^2, & q'_a &= (a - 7)^2, \\ e &\in \{0, 1\}, & D_e &= q_j - v - 1 + e, & L_e &= D_e + 1, & 0 \leq v &\leq 80, & \mu_v &= E_A^{(v)}(0) \neq 0, \\ (s_0, s_1) &= (1, 1) \text{ or } (k, j), & c'_a &= a/2 - 4, & m_a &= q_a - q'_a - v. \end{aligned} \quad (\text{CIT1})$$

The two source orders in this display remain paired throughout; write $s_k = s_0$ and $s_j = s_1$ when a degree subscript is used. All finite functionals are complex-linear. At a fixed D_e let $\mathcal{H}_0, \mathcal{H}_1$ be their original Gamma polynomial duals in degrees k, j , with full orthonormal frames

$$\phi_{r,s,c'}(S') = \frac{p_{r,s}((S' - c')/i)}{\sqrt{(2\pi)^{s/2} r! (s/2)_r}}, \quad 0 \leq r \leq D_e. \quad (\text{CIT2})$$

Use the exact CNC10 matrices $T_0 = T_{k,D_e}$, $T_1 = T_{j,D_e}$ for its original analytic numerators divided by E_A , and V_0, V_1 for every original lower evaluation. Thus $[V_0, T_0]$ has its original full column order.

2.1 The literal primitive domain and target coefficient matrix

The four multiplier matrices are completely specified without an unspecified evaluator. Let $\mathcal{B}_a$ be the original monomial-to-Gamma matrix defined by $S'^r = \sum_n (\mathcal{B}_a)_{nr} \phi_{n,s_a,c'_a}$ and set

$$\begin{aligned} U_a &= \text{diag}(1/r!)_{r=0}^{D_e} \mathcal{B}_a^\top, & (\text{Mult}_{L_e}(B_i))_{rt} &= \begin{cases} B_i^{(r-t)}(0)/(r-t)!, & r \geq t, \\ 0, & r < t, \end{cases} \\ \mathbf{T}_i &= U_j^{-1} \text{Mult}_{L_e}(B_i) U_k, & \mathbf{T} &= \text{stack}_{i=1}^4 \mathbf{T}_i, & \mathcal{V} &= \text{diag}(V_1, 4), & \mathcal{T}_1 &= \text{diag}(T_1, 4). \end{aligned} \quad (\text{CIT3})$$

Here $\text{diag}(P, 4)$ means four copies. The map U_a sends the actual native functional values to Taylor coefficients: ordinary derivatives are $\mathcal{B}_a^\top w$. This proves CIT3 and preserves every factorial and the original leading phase i^r inside $\mathcal{B}_a$. Its full inverse tuple is the analogous multiplication by the original entire D_i ; $\sum D_i B_i = 1$ gives $\mathbf{ST} = I$.

Define the raw source analytic space

$$\mathcal{S}_k^{\text{raw}} = \text{im}[V_0, T_0] \subset \mathcal{H}_0, \quad \mathcal{L}_0 = \text{im } V_0, \quad \mathcal{L}_1 = \text{im } \mathcal{V}. \quad (\text{CIT4})$$

It is exactly the set of native functionals of N/E_A for all original upper numerators N vanishing to order v . Indeed T_0 gives the original DNE quotient frame and V_0 gives exactly its allowed numerator changes by conductor multiples. The columns are independent: if $T_0 x + V_0 \eta = 0$, the CNC13 injectivity at this cutoff gives $x = 0$, and the lower Vandermonde gives $\eta = 0$. Thus $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$.

For the upper endpoint set

$$\begin{aligned} P_A &= \mathcal{T}_A \chi_j, & (\mathcal{T}_A p)(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ R_i(S') &= \mathcal{T}_{B_i} P_A(S'), & (\mathcal{T}_{B_i} p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\ \mathbf{R}_1 &= \text{stack}_{i=1}^4 (\mathcal{B}_k \text{coeff}_{S'} R_i)^\top, & \mathbf{R}_0 : \mathcal{H}_0 &\longrightarrow 0. \end{aligned} \quad (\text{CIT5})$$

All coefficient columns are padded through D_1 . Here $\deg P_A = q_j - v = D_1$ and $\text{lc } P_A = \mu_v \binom{q_j}{v}$. The complete LCC9–12 proof shows that $\mathbf{R}_1$ has rank four: the four original L_i^4 are independent on the rank-four

Segre quadric, and translation of this polynomial is injective on the 25 distinct original shifts. The native row in CIT5 is an ordinary transpose because it evaluates the complex-linear functional on R_i .

The complete primitive domain is

$$\mathcal{Y}_e^{\text{raw}} = \ker R_e, \quad \mathcal{S}_k^{\text{raw}} \subset \mathcal{Y}_e^{\text{raw}}, \quad r_e = \dim(\mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}}) = 8k + 24 - 3e. \quad (\text{CIT6})$$

For $e = 0$ it is all of $\mathcal{H}_0$. To prove the target condition, at that endpoint the raw analytic image in each $\mathcal{H}_1$ has dimension $q_j - v = L_0$ and fills it. At $e = 1$ its only missing coordinate is $\Lambda_F(P_A)$ by LCC10. Therefore Th belongs to the fourfold raw target analytic image precisely when $R_1 h = 0$. The exact numerator cochain square kills $\mathcal{S}_k^{\text{raw}}$ in those rows. Subtracting $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$ from $\dim \ker R_e = L_e - 4e$ proves CIT6.

Choose a full primitive lift matrix P_e so that $[V_0, T_0, P_e]$ is a frame of $\mathcal{Y}_e^{\text{raw}}$. This is an exact computable choice: first express CIT4–5 in ordinary Taylor coordinates, take the first nonzero pivot minors in their original order to obtain the kernel and a complement, then transport those complement columns by U_k^{-1} . It depends on the actual coefficients and imposes no metric conditioning estimate. The chosen Taylor columns are fixed; the original norm remains CIT2.

In the original target coefficient coordinates define

$$\begin{aligned} P_{\mathcal{L}_1} &= \mathcal{V}(\mathcal{V}^* \mathcal{V})^{-1} \mathcal{V}^*, \quad K_1 = (I - P_{\mathcal{L}_1}) \mathcal{T}_1, \quad \mathcal{D}_e = K_1^* K_1 = \text{diag}(D_{j,D_e}^{(s_1)}, 4), \\ F_e &= \mathcal{D}_e^{-1} K_1^* (I - P_{\mathcal{L}_1}) \mathcal{T} P_e, \quad B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}. \end{aligned} \quad (\text{CIT7})$$

The inverses exist by the full lower Vandermonde and CNC13. By CIT6 the projected columns on the right belong to $\text{im } K_1$, so the displayed left inverse gives their unique original coefficient coordinates. Consequently

$$K_1 F_e = (I - P_{\mathcal{L}_1}) \mathcal{T} P_e, \quad K_1 B_{\text{cof}} = (I - P_{\mathcal{L}_1}) \mathcal{T} T_0. \quad (\text{CIT8})$$

These are literal original finite matrices. In particular the $M_{i,k}$ retain the exact KRC scalar-bearing conjugation CNC12.

2.2 Complete connecting kernel and attained target norm

We prove the full kernel of the connecting construction. The exact finite intersection needed here is

$$\text{im } \mathcal{T} \cap \mathcal{L}_1 = \mathcal{T} \mathcal{L}_0. \quad (\text{CIT9})$$

For a direct proof, write $\text{Th} = \mathcal{V} \eta$. Apply the six original compatibility rows. Their coefficient square is $\mathcal{R} \mathcal{V} = \mathcal{V}_2 \mathcal{K}$, so $\mathcal{V}_2 \mathcal{K} \eta = 0$. At the present cutoffs $D_e + 1 - q_k = 8k + 24 - v + e \geq 16$; hence the original degree- k evaluation $\mathcal{V}_2$ is injective. The complete GKC kernel gives $\eta = \mathcal{P} \xi$. The lower root-addition square says $\mathcal{V} \mathcal{P} = \mathcal{T} V_0$, whence $\text{Th} = \mathcal{T} V_0 \xi$. Since $\mathcal{S} \mathcal{T} = I$, $h = V_0 \xi$. The reverse inclusion is exactly that same square. This proves CIT9 without a transversality or inverse-norm assumption.

For $h \in \mathcal{Y}_e^{\text{raw}}$ let $c(h)$ be the coefficient column defined as in CIT7. The native exterior differential of Th is zero, since $B_i B_t = B_t B_i$. Replacing its representative by lower evaluations changes its next differential by lower evaluations, by root addition. Therefore $c(h)$ is a cycle in the actual LCC image complex. The complete map is

$$\begin{aligned} \partial_e : \mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}} &\longrightarrow H^1(\mathcal{I}_e), \quad [h] \longmapsto [c(h)], \\ \text{im } \partial_e &= \mathcal{C}_e, \quad \mathcal{C}_e = \mathcal{S}_e/\mathcal{B}, \quad \mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e], \quad \mathcal{B} = \text{im } B_{\text{cof}}. \end{aligned} \quad (\text{CIT10})$$

If $h = T_0 x + V_0 \xi$, its target coefficients are the boundary $B_{\text{cof}} x$ by CIT8. Conversely, if $c(h) = B_{\text{cof}} x$, then $\mathcal{T}(h - T_0 x) \in \mathcal{L}_1$ and CIT9 gives $h - T_0 x \in \mathcal{L}_0$. Thus the kernel before quotient is exactly $\mathcal{S}_k^{\text{raw}}$. This proves injectivity and the stated presentation; $[B_{\text{cof}}, F_e]$ is independent. It is exactly the positive connecting image of $H^0(\mathcal{R}_e) = \ker \mathfrak{t}_e$ in LCC13, since its construction lifts the original degree-zero quotient cycle and takes its differential with the same positive sign. No splitting of the remaining $\ker U_e$ has been introduced.

Its full cohomology restriction Gram in the P_e image coordinates is

$$\begin{aligned} H_{\text{conn},e} &= F_e^* \mathcal{D}_e F_e \\ &\quad - F_e^* \mathcal{D}_e B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_e B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_e F_e. \end{aligned} \quad (\text{CIT11})$$

The exact least norm over every original boundary gives this Schur complement. All original lower coefficients were already minimized simultaneously in $\mathcal{D}_e$. Equivalently set

$$Z_e^{\text{raw}} = [\mathcal{V}, \mathbb{T}T_0], \quad H_{\text{conn},e} = (\mathbb{T}P_e)^*(I - P_{Z_e^{\text{raw}}})\mathbb{T}P_e, \quad P_Z = Z(Z^*Z)^{-1}Z^*. \quad (\text{CIT12})$$

The columns of Z_e^{raw} are independent by CIT9 and the independence in CIT4. The same argument with the added P_e proves independence of $[Z_e^{\text{raw}}, \mathbb{T}P_e]$. Indeed the complete minimization in CIT11 allows exactly $\mathbb{T}T_0x + \mathcal{V}\eta$. This proves CIT12 and positivity. It also gives the full original raw determinant

$$\det H_{\text{conn},e} = \frac{\det([Z_e^{\text{raw}}, \mathbb{T}P_e]^*[Z_e^{\text{raw}}, \mathbb{T}P_e])}{\det((Z_e^{\text{raw}})^*Z_e^{\text{raw}})}. \quad (\text{CIT13})$$

These are the same fixed combined frames, so their ordinary Schur determinant introduces no missing coordinate factor.

2.3 The original source-to-image determinant at its low cutoff

Give the primitive source quotient its original source metric and its correlated image metric, respectively:

$$H_{\text{src},e} = P_e^*(I - P_{[V_0, T_0]})P_e, \quad H_{\text{cor},e} = (\mathbb{T}P_e)^*(I - P_{\mathbb{T}[V_0, T_0]})\mathbb{T}P_e. \quad (\text{CIT14})$$

These are two explicit attained minima. The original NBG $\|\mathbb{T}\| \leq K_+$ and $\|\mathbb{S}\| \leq K_-$ with $\mathbb{S}\mathbb{T} = I$ imply

$$K_-^{-2}H_{\text{src},e} \preceq H_{\text{cor},e} \preceq K_+^2H_{\text{src},e}. \quad (\text{CIT15})$$

For each fixed primitive value, apply the two norm inequalities to every representative in $P_e b + \mathcal{S}_k^{\text{raw}}$ and take the minima. This proves both directions on the exact same fibre. The finite constants are the complete NBG12 ones at base k and cutoff D_e , including source order $(1, 1)$ or (k, j) and center shift $(\beta_{4,a,b} - 2)/i$.

There is also an exact independent-gauge quotient identity. In the original source quotient put

$$X_k = \mathcal{S}_k^{\text{raw}}/\mathcal{L}_0, \quad Y_e = \mathcal{Y}_e^{\text{raw}}/\mathcal{L}_0, \quad X_k \subset Y_e.$$

Their correlated target images are the original NKC9 images. The original independent-gauge form restricted to Y_e and then quotiented by X_k allows precisely the total fibre $\mathbb{T}T_0x + \mathcal{L}_1$; its correlated quotient allows $\mathbb{T}T_0x + \mathbb{T}\mathcal{L}_0$. These are CIT12 and CIT14. Hence the literal PCJ7 source quotient gives

$$\frac{\det H_{\text{conn},e}}{\det H_{\text{cor},e}} = \det \Omega_{Y_e/X_k, D_e}. \quad (\text{CIT16})$$

If R transports the original primitive lifts to an adapted correlated quotient frame, both Grams are changed by $R^*(\cdot)R$, and their two factors $|\det R|^2$ cancel in CIT16. This is the exact coordinate map behind that determinant equality.

Use the actual compatibility-plus-complement expressions χ_{X_k, D_e} and χ_{Y_e, D_e} of PCJ1. The complete shared-update proof PCJ13–16 gives

$$\begin{aligned} \log \det H_{\text{conn},e} - \log \det H_{\text{src},e} &= -(\chi_{X_k, D_e} - \chi_{Y_e, D_e}) + \epsilon_e, \\ -2r_e \log \max(1, K_-) - r_e \log c_+ &\leq \epsilon_e \leq 2r_e \log \max(1, K_+) - r_e \log c_-. \end{aligned} \quad (\text{CIT17})$$

Indeed write the left side as the correlated/source logarithm plus $\log \det \Omega_{Y_e/X_k}$. The first is bounded by CIT15. PCJ16 writes the negative of the latter as $\chi_{X_k} - \chi_{Y_e} + \nu$, with $r_e \log c_- \leq \nu \leq r_e \log c_+$. Subtraction gives exactly CIT17. Here $c_- = \min(1, L_-^{-2})$ and $c_+ = \max(1, L_+^2)$ are the original NKC constants at this cutoff. Consequently this specified error is $O_{\text{actual}}(k \log q_k) = o(kq_k)$ in both original source pairs. At $e = 0$, Y_0 is the full ambient source quotient, so its source-complement projector vanishes and $\chi_{Y_0} = 0$ exactly. At $e = 1$ its four actual polynomial constraints remain; χ_{Y_1} is not set to zero.

CIT14–17 concern the defining low cutoff D_e . They do not assert that a primitive truncated germ has a common compatible extension to every later cutoff. The target coefficient columns F_e now are fixed, and their complete later metrics are calculated next without that assertion.

2.4 The fixed connecting flag at every original target row

For $q_j - 1 \leq N \leq 2q_j$ put $D = N - v$ and retain the fixed B_{cof}, F_e in the original target coefficients. Define

$$\begin{aligned} \mathcal{D}_N &= \text{diag}(D_{j,N-v}^{(s_1)}, 4), & Z_{B,N} &= [\mathcal{V}_N, \mathcal{T}_{1,N} B_{\text{cof}}], \\ Z_{C,N} &= [Z_{B,N}, \mathcal{T}_{1,N} F_e], & H_{\text{conn},e}(N) &= \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e / \mathcal{B}), \\ \det H_{\text{conn},e}(N) &= \det(Z_{C,N}^* Z_{C,N}) / \det(Z_{B,N}^* Z_{B,N}). \end{aligned} \quad (\text{CIT18})$$

The original DNE positivity and the independence in CIT10 make both raw frames full column rank at every one of these indices. Eliminating the complete lower frame and then B_{cof} proves the determinant identity. At the defining $N = q_j - 1 + e$, it is exactly CIT11–13 by CIT8.

Every entry is given by the original arithmetic recursion. If $\nu_{a,n} = \sum (Z_{\text{exp},j})_{uv,a} \beta_{j,u,v}^n$, then

$$\begin{aligned} f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1} (b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \\ t_n &= \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} \beta_{j-8,u,v}^m}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,v}. \end{aligned} \quad (\text{CIT19})$$

Arrays outside their polynomial range are zero. Differentiating $E_A F_a = N_a$ gives the first line; substituting the full original Gamma recurrence gives the next. Thus these are exactly the unchanged IRR7–9 rows, retaining all original moments and phases. The new four-row blocks in CIT18 are $z_{B,n} = [\text{diag}(v_n, 4), \text{diag}(t_n, 4) B_{\text{cof}}]$ and $z_{C,n} = [z_{B,n}, \text{diag}(t_n, 4) F_e]$. For $X = B, C$ define $G_{X,n} = Z_{X,n+v}^* Z_{X,n+v}$. In degree-major row order each raw Gram increases by $z^* z$. The fixed row permutation from copy-major order is unitary.

For $q_j - v \leq n \leq 2q_j - v$, the exact connecting-volume increment is

$$\log \frac{\det H_{\text{conn},e}(n+v)}{\det H_{\text{conn},e}(n+v-1)} = \log \det(I_4 + z_{C,n} G_{C,n-1}^{-1} z_{C,n}^*) - \log \det(I_4 + z_{B,n} G_{B,n-1}^{-1} z_{B,n}^*) \geq 0. \quad (\text{CIT20})$$

The determinant lemma proves the equality. For its sign, transform the old extra columns to their full least-norm section over $Z_{B,n+v-1}$. The larger row covariance is the smaller one plus $RQ^{-1}R^* \succeq 0$, by block inversion. This proves the inequality and retains the actual old quotient denominator. No independent rowwise gauge minimum is taken.

2.5 Its exact place in the common low invariant loss

Let $\mathbb{J} = \ker \mathbb{B}$ be the full original fourfold DNE invariant kernel. At each later N use the same actual target metric $\mathcal{D}_N$ and fixed flags

$$\begin{aligned} W &= \mathcal{B} + \mathbb{J}, & K_{\text{conn},e} &= \mathcal{S}_e \cap W = \mathcal{B} + (\mathcal{S}_e \cap \mathbb{J}), \\ T_{\text{conn},e} &= \mathcal{S}_e + W = \mathcal{S}_e + \mathbb{J}, \\ \omega_{\text{conn},e}(N) &= \frac{\det \text{Gram}_{\mathcal{D}_N}(T_{\text{conn},e}/W)}{\det \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/K_{\text{conn},e})}. \end{aligned} \quad (\text{CIT21})$$

The two Grams use the same quotient-value coordinates under $[z] \mapsto [\mathbb{B}z]$. Its original fibres are exactly the two spaces in the denominator and numerator, as in LCC20–23. The source is the corresponding quotient of the connecting cohomology image; its full kernel is $(\mathcal{S}_e \cap \mathbb{J})/(\mathcal{B} \cap \mathbb{J})$. Thus CIT21 is its actual further target loss, and is in $(0, 1]$.

The full low cycles contain $\mathcal{S}_e$, and LCC13 gives

$$\dim(\mathcal{L}_e / \mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0). \quad (\text{CIT22})$$

Both spaces are fixed original coefficient flags for the later metric comparisons. The difference of their invariant loss returns has the finite bound

$$|\mathcal{R} \log \omega_e - \mathcal{R} \log \omega_{\text{conn},e}| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \quad (\text{CIT23})$$

Here $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$. For a complete verification, apply the actual inclusions $\mathcal{S}_e \subset \mathcal{Z}_e$ with their common W . The kernel/intersection quotient and sum quotient fit into the exact sequence $0 \rightarrow K_{\text{full}}/K_{\text{conn}} \rightarrow \mathcal{Z}_e/\mathcal{S}_e \rightarrow T_{\text{full}}/T_{\text{conn}} \rightarrow 0$ by the identity on representatives. Their two ranks sum to κ_e . The full adapted-Gram factorization LFC7–12 expresses the loss difference as the sum of those two quotient log determinants minus the $\mathcal{Z}_e/\mathcal{S}_e$ log determinant and a fixed frame constant. FRE’s same native form sandwich puts the three monotone returns between zero and twice their ranks times $\log A_{j,s_1}^{\text{end}}$. Both sides of the resulting signed difference are bounded by $2\kappa_e \log A$. This proves CIT23, including $\kappa_e = 0$, with all frame constants canceled only after the original four signs.

There is a further exact relation to the ORIGINAL ITL source. Write $F_{\text{inv}} = \text{stack}_i(M_{i,k}J_0)$. Since $B_{\text{cof}} = \text{stack}_i M_{i,k}$,

$$\text{im } F_{\text{inv}} \subset \mathcal{B}, \quad \mathbb{J} \subset W_{\text{inv}} := \mathbb{J} + \text{im } F_{\text{inv}} \subset W \subset T_{\text{conn},e}. \quad (\text{CIT24})$$

These inclusions are literal original coefficient maps. The ITL source therefore represents boundaries in low cohomology, but its target Gram remains in the denominator $\text{Gram}(W)$ of the cohomology target quotient.

To exhibit every factor, let $I_R = F_{\text{inv}}Z_R$ be exactly the fixed ITL4 source-kernel complement and choose fixed columns R_b completing $[X_{\mathbb{J}}, I_R]$ to W . In a specified fixed full frame X_W , write

$$[X_{\mathbb{J}}, I_R, R_b] = X_W C_W, \quad Q_b(N) = \text{Gram}_{\mathcal{D}_N}(W/W_{\text{inv}}) \quad \text{in the lifts } R_b.$$

The first quotient of this chain has exactly ITL’s target Gram $H_{\text{inv}}(N)$: its fibre is $I_R b + \mathbb{J}$. Its source pullback quotient is the full ITL $A_{\text{inv}}(N)$, and $\det H_{\text{inv}} = \det A_{\text{inv}} \omega_{\text{inv}}$. Sequential Schur complements consequently give

$$|\det C_W|^2 g_W(N) = g_{\mathbb{J}}(N) \det H_{\text{inv}}(N) \det Q_b(N), \quad g_X(N) = \det(X_X^* \mathcal{D}_N X_X). \quad (\text{CIT25})$$

All matrices in this formula are in their stated independent frames; $r_{\text{inv}} = 0$ gives the empty first quotient.

Finally retain the full LFC coordinate constant $c_{\text{conn},e}$ from the adapted frames for

$$(\mathcal{S}_e, K_{\text{conn},e}, T_{\text{conn},e}, W).$$

Its cross ratio and CIT25 give the exact original identity

$$\begin{aligned} \log \omega_{\text{conn},e} + \log \omega_{\text{inv}} &= \log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{CIT26})$$

This relates the two actual losses through their exact common boundary structure; it does not identify their source spaces. Taking the original four-return cancels the last two fixed coordinate constants. Equivalently ITL16’s exact matched relative operator satisfies $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, where Q_{inv} is the full original source quotient $J_0^* D_{k,N-v}^{(s_0)} J_0 / E$. Substituting this equality instead in CIT25 replaces $(\omega_{\text{inv}}, A_{\text{inv}})$ in CIT26 by $(\det \mathcal{R}_X, Q_{\text{inv}})$, preserving the source-order pair $(1, 1)$ or (k, j) . It retains the original source mass.

Thus the surviving connecting target is given by the explicit matrix CIT7, its complete target minimum CIT11–13, the original low source-to-image determinant CIT17, and all later original row formulas CIT18–20. The fixed-rank remainder is controlled by CIT23. The remaining common target loss has the exact original ITL boundary relation CIT24–26, with its whole residual overlap, all absolute native Grams and both source measures still present.

3 Primitive-source transport inside the original connecting image

Keep the complete original PDS1–24 and CIT1–26 data at fixed $k = 4l + 1 \geq 9$, $j = k + 4$, $D_e = (k + 5)^2 - v - 1 + e$, and either original source-order pair $(s_0, s_1) = (1, 1)$ or (k, j) . All polynomial coefficients, period data, lower gauges and exterior signs remain those of those constructions. In this section $P_e, F_e, T_{a,D_e}, V_{a,D_e}$ are CIT’s primitive lifts, connecting target columns, analytic frame and full lower frame. The polynomial coefficient matrix C_e and metric $G_e = \overline{(C_e^* C_e)^{-1}}$ are PDS7. The boundary column matrix is the unchanged $B_{\text{cof}} = \text{stack}_i M_{i,k}$.

3.1 The exact source coordinate map and its target naturality

The coordinate maps between the two proved primitive presentations are

$$\Theta_e = C_e^\top P_e, \quad \Theta_0 : \mathbb{C}^{d_0} \xrightarrow{\simeq} \mathbb{C}^{d_0}, \quad \Theta_1 : \mathbb{C}^{d_0-3} \xrightarrow{\simeq} \ker W^\top, \quad d_0 = 8k + 24. \quad (\text{PDI1})$$

The inverse signs mean inverses onto the displayed images. Indeed PDS4 is the onto quotient by precisely CIT4's analytic source, and P_e is a full basis of that primitive quotient. The PDS8 four-row condition is CIT5's same polynomial differential. Thus both maps are bijections onto the asserted spaces. Equality of the exact minimum-norm fibres proves

$$H_{\text{src},e} = \Theta_e^* G_e \Theta_e. \quad (\text{PDI2})$$

For completeness a native vector w and $P_e a$ have the same PDS quotient coordinates exactly when $C_e^\top(w - P_e a) = 0$, equivalently $w - P_e a \in \text{im}[V_{k,D_e}, T_{k,D_e}]$ by PDS4. The minimum over this fibre is both sides of PDI2, by CIT14 and PDS7. This proves the metric identity as well as its linear map.

Let $J_k^{\text{tr}} : \mathcal{H}_{k,D_1} \rightarrow \mathcal{H}_{k,D_0}$ and $J_j^{\text{tr}} : \mathcal{H}_{j,D_1} \rightarrow \mathcal{H}_{j,D_0}$ be literal jet truncation, so they delete the last native functional coordinate; the source order and centre are fixed during this truncation. Let pr_0 delete the last polynomial quotient coordinate as in PDS11, and set

$$Z = \text{pr}_0 \Theta_1, \quad R = \Theta_0^{-1} Z, \quad \text{rank } R = d_0 - 3, \quad \Theta_1 = EZ. \quad (\text{PDI3})$$

The rank assertion and last equality follow from PDS11. The first d_0 relation polynomials are unchanged between cutoffs; therefore $C_0^\top J_k^{\text{tr}} P_1 = \text{pr}_0 \Theta_1 = \Theta_0 R$. Their difference is in the old analytic source. In its original independent frame there are unique matrices L and A such that

$$J_k^{\text{tr}} P_1 = P_0 R + V_{k,D_0} L + T_{k,D_0} A. \quad (\text{PDI4})$$

They are explicit finite matrices: with $U = [V_{k,D_0}, T_{k,D_0}]$ their exact block expression is

$$\begin{bmatrix} L \\ A \end{bmatrix} = (U^* U)^{-1} U^* (J_k^{\text{tr}} P_1 - P_0 R).$$

The actual connecting coefficient columns satisfy

$$\boxed{F_1 = F_0 R + B_{\text{cof}} A.} \quad (\text{PDI5})$$

To prove it, the original multiplier square commutes with truncation: $(\text{diag}(J_j^{\text{tr}}, 4)) \Upsilon^{(1)} = \Upsilon^{(0)} J_k^{\text{tr}}$. This is the equality of the Taylor convolutions in CIT3 after discarding the last coefficient. Likewise truncation takes each original lower or analytic target column to the corresponding old column, because they are jets of the same entire or analytic germ. Apply these equalities to the target representative of F_1 in CIT8 and use PDI4. Its old target jet differs from that of $F_0 R + B_{\text{cof}} A$ by an allowed lower evaluation, using the root-addition square on $V_{k,D_0} L$. The original degree- j analytic coefficient map modulo lower evaluations is injective at D_0 by CNC13. Thus the coefficient columns themselves agree, proving PDI5. No inverse estimate is used to pass from equality of the jets to these coefficients.

In particular, in the single original target coefficient space,

$$\mathcal{S}_1 = \text{im}[B_{\text{cof}}, F_1] \subset \mathcal{S}_0 = \text{im}[B_{\text{cof}}, F_0], \quad \dim(\mathcal{S}_0/\mathcal{S}_1) = 3. \quad (\text{PDI6})$$

Independence of each displayed frame is proved in CIT10, and PDI3 gives the rank of the quotient inclusion. This proves the codimension and identifies the actual map from primitive truncation to the fixed connecting coefficient flags.

3.2 The two low metric increments in common coordinates

The old source restricted to this image and the upper source, in precisely the P_1 coordinates, have Grams

$$K_{\text{src},0} = R^* H_{\text{src},0} R = Z^* G_0 Z, \quad K_{\text{src},1} = H_{\text{src},1} = (EZ)^* G_1 (EZ). \quad (\text{PDI7})$$

PDS12–24 now apply to these actual connecting-source frames:

$$K_{\text{src},1} = K_{\text{src},0} + \frac{(\eta Z)^*(\eta Z)}{\sigma}, \quad 0 \leq \Delta_{\text{src}} := \log \frac{\det K_{\text{src},1}}{\det K_{\text{src},0}} \leq \log U_* = o(kq_k). \quad (\text{PDI8})$$

Every constant in U_* is the original PDS19 finite product; it has the fully proved explicit bound PDS24.

In the original target define

$$K_{\text{tgt},0} = R^* H_{\text{conn},0} R, \quad K_{\text{tgt},1} = H_{\text{conn},1}. \quad (\text{PDI9})$$

By PDI5 these are the metrics of the same fixed quotient $\mathcal{S}_1/\mathcal{B}$ at the two native target cutoffs, in the same F_1 value coordinates. Adding the boundary $B_{\text{cof}}A$ does not change its minimum over that boundary. This proves the first identity in PDI9 as a quotient statement, including its frame, rather than presuming equality of two arbitrary representatives.

Here is its complete four-row update. At D_0 set

$$\begin{aligned} N_B &= [\text{diag}(V_{j,D_0}, 4), \text{diag}(T_{j,D_0}, 4)B_{\text{cof}}], & N_F &= \text{diag}(T_{j,D_0}, 4)F_1, \\ G_B &= N_B^* N_B, & G_{BF} &= N_B^* N_F, & K_{\text{tgt},0} &= N_F^* N_F - G_{BF}^* G_B^{-1} G_{BF}. \end{aligned} \quad (\text{PDI10})$$

Both original lower and boundary frames are kept in N_B . Let z_B, z_F be the next degree's four rows of these very frames, computed from CIT19, and define

$$S_B = I_4 + z_B G_B^{-1} z_B^*, \quad r_B = z_F - z_B G_B^{-1} G_{BF}. \quad (\text{PDI11})$$

Their exact update is

$$\begin{aligned} K_{\text{tgt},1} &= K_{\text{tgt},0} + r_B^* S_B^{-1} r_B, \\ \Delta_{\text{tgt}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{tgt},0}} = \log \det \left(I_4 + S_B^{-1/2} r_B K_{\text{tgt},0}^{-1} r_B^* S_B^{-1/2} \right) \geq 0. \end{aligned} \quad (\text{PDI12})$$

For proof replace N_F by its old orthogonal residual $N_F - N_B G_B^{-1} G_{BF}$. In this unchanged quotient the old joint Gram is $\text{diag}(G_B, K_{\text{tgt},0})$; the added rows are $[z_B, r_B]$. The new Schur complement is $K_{\text{tgt},0} + r_B^* [I - z_B (G_B + z_B^* z_B)^{-1} z_B^*] r_B$. Multiplying by $I + z_B G_B^{-1} z_B^*$ proves that the bracket is S_B^{-1} . Taking the determinant gives PDI12. The column operation has identity diagonal blocks, so its determinant is one and it leaves no missing frame factor.

There is a finite bound using the complete original native next-row budget. Write $D_0 = D_{j,D_0}^{(s_1)}$ for the single-copy original coefficient metric, $T = T_{j,D_0}$, $V = V_{j,D_0}$, and let t, v_{row} be their next original rows. Set

$$\begin{aligned} A_V &= V^* V, \quad \zeta = v_{\text{row}} A_V^{-1} v_{\text{row}}^*, \quad \rho = t - v_{\text{row}} A_V^{-1} V^* T, \\ \beta_{\text{next}} &= 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right), \quad 0 \leq \Delta_{\text{tgt}} \leq \beta_{\text{next}} \leq 4 \log A_{j,s_1}^{\text{end}} = o(kq_k). \end{aligned} \quad (\text{PDI13})$$

The exact lower-gauge Schur update is $D_1 = D_0 + \rho^* \rho / (1 + \zeta)$, as in IRR and LFC. Thus the full four-copy ambient log-determinant increment is β_{next} . To prove the middle bound without multiplying by a quotient dimension, put the old ambient form into its isometric coordinates and write the new one as e^B with $B \succeq 0$. For any fixed subspace with full frame Y , differentiation along e^{tB} gives

$$\frac{d}{dt} \log \det(Y^* e^{tB} Y) = \text{Tr}(B P_Y(t)), \quad P_Y(t) = e^{tB/2} Y (Y^* e^{tB} Y)^{-1} Y^* e^{tB/2}.$$

$P_Y(t)$ is the orthogonal projector onto $e^{tB/2} Y$. For nested fixed spaces its two projectors are ordered, so the derivative of their quotient log determinant is between zero and $\text{Tr } B$. Integrating proves that the increment of $\mathcal{S}_1/\mathcal{B}$ is at most the full ambient increment, which is β_{next} . Finally FRE's original endpoint form bound implies that each of the four identical single-copy rank-one relative eigenvalues is at most A_{j,s_1}^{end} , proving the last inequality. Its already proved growth is $\log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k)$. The isometric coordinates in this proof establish a comparison of the unchanged forms; the formulas PDI10–13 remain in their original frames.

The exact adjacent source-to-image determinant correction is therefore the difference of the explicit four-row and one-row log determinants PDI12 and PDI8:

$$\begin{aligned} \Delta_{\text{image/source}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{src},1}} - \log \frac{\det K_{\text{tgt},0}}{\det K_{\text{src},0}} = \Delta_{\text{tgt}} - \Delta_{\text{src}}, \\ -\log U_* &\leq \Delta_{\text{image/source}} \leq \beta_{\text{next}}, \quad \Delta_{\text{image/source}} = o(kq_k). \end{aligned} \quad (\text{PDI14})$$

The lower determinant uses the proved common primitive subspace R ; its three-dimensional complement has not been silently discarded from the full old cohomology.

3.3 The two fixed connecting invariant losses

At every later original target cutoff keep the literal flags of CIT21 and the same $W = \mathcal{B} + \mathbb{J}$. By PDI6 their nesting has codimension exactly three. The identity on representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow \mathcal{S}_0/\mathcal{S}_1 \longrightarrow T_0/T_1 \longrightarrow 0, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \quad (\text{PDI15})$$

Surjectivity follows from the definition of the sum; its kernel consists of the class of $x \in \mathcal{S}_0$ with $x \in \mathcal{S}_1 + W$, hence of $K_0 + \mathcal{S}_1$ modulo $\mathcal{S}_1$. Its kernel presentation is K_0/K_1 . Thus the two end ranks sum to three. The complete adapted Gram factorization LFC7–12 expresses the difference of the two loss logarithms as $\log \det \text{Gram}(T_0/T_1) + \log \det \text{Gram}(K_0/K_1) - \log \det \text{Gram}(\mathcal{S}_0/\mathcal{S}_1)$ plus a fixed coordinate constant. Under the original four-return $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$ that constant cancels. FRE bounds each monotone quotient return between zero and twice its rank times $\log A_{j,s_1}^{\text{end}}$. The positive two ranks sum to three and the negative rank is three; therefore

$$|\mathcal{R} \log \omega_{\text{conn},0} - \mathcal{R} \log \omega_{\text{conn},1}| \leq 6 \log A_{j,s_1}^{\text{end}} = o(kq_k). \quad (\text{PDI16})$$

This is the complete primitive-image refinement of the full-cohomology comparison in LFC. The retained $\ker U_e$ constituents of the full LCC sequence are still present, with their explicit CIT23 bounds. The common long connecting loss continues to be the exact target expression CIT21 and its ITL boundary factorization CIT24–26. The original low source transport now feeds that target calculation through the explicit coefficient identity PDI5, the two native increments PDI12–14, and the proved fixed-flag comparison PDI16.

4 The original monic source band and its low-polynomial quotient

This calculation evaluates the source band in PDS4 before its actual conductor translation. The metric distortion of that translation is treated separately and is not assumed here.

4.1 Original coordinates and finite parameters

Fix the actual simple quartet, period, branch and conductor of PDS. Write $\sigma = (\delta^2 + \gamma^2)^{1/2}$, with $\delta, \gamma > 0$, and retain

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad q = (k + 1)^2, \quad e \in \{0, 1\}, \quad d = 8k + 24 + e, \\ \nu &= q + d - 1 = (k + 5)^2 - 1 + e, \quad c_0 = k/2, \quad c_1 = k/2 - 4, \\ \beta_{k,a,b} &= k(1/2 - \delta - i\gamma) + 2a\delta + 2bi\gamma, \quad \chi_k(S) = \prod_{a,b=0}^k (S - \beta_{k,a,b}), \\ u_a(S) &= \chi_k(S)S^a \quad (0 \leq a < d), \quad s = 1 \text{ or } k, \quad \ell_s = (s - 1)/4, \quad Q_s = q + \ell_s, \\ \theta &= \pi/2, \quad h = 1/q, \quad \kappa = \theta - h, \quad R = k\sigma, \quad B = \max(1, R, k/2), \quad m = 2Q_s/\theta. \end{aligned} \quad (\text{PSG1})$$

The moment exponent Q_s is distinct from the target root count $(k+5)^2$. The complete input norm and polynomials are

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \quad h_{n,s} = M_s n! (s/2)_n, \\ p_0 &= 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(n-1 + s/2)p_{n-1}, \quad \phi_n(S) = p_n((S - c_0)/i)/\sqrt{h_{n,s}}, \\ \|p\|_{s,c_0}^2 &= \int_{\mathbb{R}} |p(c_0 + iy)|^2 dm_s(y), \quad H_{e,s}^\chi = \text{Gram}_{s,c_0}(u_0, \dots, u_{d-1}). \end{aligned} \quad (\text{PSG2})$$

These are PDS6's full Gamma norms with the specified input center; the output of translation retains c_1 . The original roots in the y coordinate satisfy

$$\zeta_{a,b} = (\beta_{k,a,b} - c_0)/i = (2b - k)\gamma - i(2a - k)\delta, \quad |\zeta_{a,b}| \leq R, \quad \chi_k(c_0 + iy) = i^q \prod_{a,b} (y - \zeta_{a,b}). \quad (\text{PSG3})$$

The change from columns $\chi_k S^a$ to $\chi_k((S - c_0)/i)^a$ is triangular with diagonal i^{-a} . Its determinant has modulus one, proving equality of the Gram determinants while retaining the entire coordinate change.

4.2 The complete Gamma density

Set

$$\beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad L_h = C_L(\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi}(\sin h)^{-1/2}. \quad (\text{PSG4})$$

Applying the Gamma recurrence ℓ_s times gives exactly

$$dm_s(y) = \beta_s \prod_{r=0}^{\ell_s-1} (y^2 + (2r + 1/2)^2) d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{PSG5})$$

The recurrence contributes $4^{-\ell_s}$, cancelling $2^{2\ell_s}$ from $2^{s/2}$; the remaining scalar is $\beta_s/(2\pi)$. Empty products are one.

Rotation of the Euler integral through $\text{sgn}(y)(\pi/2 - h)$, for $a > 0$, gives

$$|\Gamma(a + iy/2)| \leq \Gamma(a)(\sin h)^{-a} e^{-(\pi/2 - h)|y|/2}.$$

For justification, truncate between radii ε, T . The small arc is $O(\varepsilon^a)$; the large arc vanishes because $\cos t \geq \sin h > 0$ throughout the sector. The absolute ray integral is $\int_0^\infty e^{-r \sin h} r^{a-1} dr = \Gamma(a)(\sin h)^{-a}$. The exact reflection product is

$$|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2 / \cosh(\pi y).$$

Dividing by the upper bound for its second Gamma factor and using $\cosh(\pi y) \leq e^{\pi|y|}$ proves

$$L_h e^{-(\theta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\theta-h)|y|} \quad (y \in \mathbb{R}). \quad (\text{PSG6})$$

This includes zero and the full inner region; PSG5 retains the mass.

4.3 Finite determinant bounds in all original roots

Define

$$f_\theta(x) = 2x \log(2x/\theta) - 2x \quad (x > 0). \quad (\text{PSG7})$$

Whenever $m \geq 2R$, one has the explicit inequalities

$$\begin{aligned} d \log \beta_s + df_\theta(Q_s) - B_{k,e,s}^- &\leq \log \det H_{e,s}^\chi \\ &\leq d \log \beta_s + df_\theta(Q_s) + B_{k,e,s}^+, \end{aligned} \quad (\text{PSG8})$$

with

$$\begin{aligned} B_{k,e,s}^- &= d \log^+(1/L_h) + \theta^2 d^3 / Q_s + dh(m+2d) + 4dqR/m, \\ B_{k,e,s}^+ &= d \log^+ U_h + d\kappa B + d \log 2 + d(1 + \log(2Q_s)) \\ &\quad + d(d-1) \log(2Q_s + 2d) + d(2Q_s + d) \log(\theta/\kappa). \end{aligned} \quad (\text{PSG9})$$

Here $\log^+ x = \max(0, \log x)$. For example the finite condition holds for $k \geq \max(9, \lceil \theta \sigma \rceil + 1)$, since $Q_s \geq q > k^2$.

For the upper bound, each original root factor at y or $-y$, $y \geq 0$, is at most $y + R$ in modulus. Each positive factor in PSG5 is at most $(y + B)^2$. The a th column in the y frame therefore has squared norm at most

$$2\beta_s U_h e^{\kappa B} \frac{(2Q_s + 2a)!}{\kappa^{2Q_s + 2a + 1}}. \quad (\text{PSG10})$$

Indeed $y^{2a} \leq (y + B)^{2a}$; substitute $t = y + B$ and enlarge the remaining integral from $[B, \infty)$ to $[0, \infty)$. Hadamard's inequality gives the product of these d bounds. In this setting its proof is the successive orthogonal projection of the columns: every squared remainder norm is at most its original norm, and their product is the Gram determinant.

Monotonicity of $\log x$ gives, for every integer $n \geq 1$,

$$n \log n - n + 1 \leq \log(n!) \leq n \log n - n + 1 + \log n. \quad (\text{PSG11})$$

Also $(2Q_s + 2a)! \leq (2Q_s)!(2Q_s + 2d)^{2a}$. In the product of PSG10 the total extra factorial exponent is $\sum_a 2a = d(d-1)$, and the rate exponent is $\sum_a (2Q_s + 2a + 1) = d(2Q_s + d)$. Substituting PSG11 proves the upper side of PSG8. The remaining term $-d^2 \log \theta$ is nonpositive since $\theta = \pi/2 > 1$; it is discarded only in this upper inequality, not in PSG10.

For the lower bound, write $w(y)dy = |\chi_k(c_0 + iy)|^2 dm_s(y)$. Expansion of two Vandermonde determinants and Fubini gives the exact identity

$$\det H_{e,s}^X = \frac{1}{d!} \int_{\mathbb{R}^d} \prod_{i < t} (y_t - y_i)^2 \prod_{i=1}^d w(y_i) dy_i. \quad (\text{PSG12})$$

Every moment converges by PSG6; the $d!$ permutations of one Vandermonde produce the same moment determinant. Retain the d original-coordinate intervals $I_i = [m + 2i, m + 2i + 1]$, $0 \leq i < d$. On their product every pair distance is at least one. Its $d!$ permutations are disjoint and cancel $1/d!$. All intervals have length one.

If $y = m + t$ lies in their union then $0 \leq t \leq 2d$. The inequality $\log(1+x) \geq x - x^2/2$, $x \geq 0$, gives

$$2Q_s \log y - \theta y \geq f_\theta(Q_s) - \theta^2 d^2 / Q_s.$$

Also $|y - \zeta| \geq y - R$, and $R/y \leq 1/2$. Integrating $1/(1-x) \leq 2$ on this interval yields

$$|\chi_k(c_0 + iy)|^2 \geq y^{2q} e^{-4qR/y} \geq y^{2q} e^{-4qR/m}.$$

The recurrence factors in PSG5 are at least $y^{2\ell_s}$. Thus throughout the retained intervals

$$w(y) \geq \beta_s L_h \exp(f_\theta(Q_s) - \theta^2 d^2 / Q_s - h(m + 2d) - 4qR/m).$$

Substitution into PSG12 proves the lower side of PSG8. For fixed actual data, $Q_s \asymp q$, $d = O(k)$, $R, B = O(k)$, and $h = 1/q$. Using $2h/\pi \leq \sin h \leq h$ and $\log(\theta/\kappa) \leq h/(\theta - h)$ in PSG9 proves

$$B_{k,e,s}^\pm = O_{\text{actual}}(q \log q) = o(kq). \quad (\text{PSG13})$$

The bounds themselves remain the finite expressions PSG9.

4.4 The exact original low-polynomial quotient

Keep $v = \text{ord}_0 E_A \leq 80$, $\mu_v \neq 0$, and let

$$U_e = \text{span}(u_0, \dots, u_{d-1}), \quad L_v = \mathbb{C}[S]_{<v}, \quad J_{e,s}^\chi = \text{Gram}((U_e + L_v)/L_v) \text{ in the lifts } u_a. \quad (\text{PSG14})$$

A nonzero multiple of χ_k has degree at least $q > v$; therefore $U_e \cap L_v = 0$ and this quotient Gram is positive. For $v = 0$, $J^\chi = H^\chi$ and set $C_E = 1$. For $v \geq 1$, $k \geq v - 1$, choose the original nodes

$$z_b = \zeta_{0,b} = (2b - k)\gamma + ik\delta \quad (0 \leq b < v), \quad Z = \max_{b < v} |z_b|, \quad A_{v,s} = \max(1, 2(v + 1 + s/2)).$$

Their differences are exactly $2\gamma(b - t)$. Define

$$\begin{aligned} (C_L^{\text{int}})^2 &= \sum_{b=0}^{v-1} \frac{\prod_{t \neq b} (A_{v,s} + |z_t|)^2}{(2\gamma)^{2(v-1)} (b!(v-1-b)!)^2}, \\ \rho &= 1/2, \quad a_\rho = \text{arctanh } \rho, \\ C_E^2 &= \max \left\{ 1, v(C_L^{\text{int}})^2 (\nu + 1) \max(1, 2\sqrt{\nu}) (1 - \rho^2)^{-s/2} e^{2Za_\rho} \rho^{-2\nu} \right\}. \end{aligned} \quad (\text{PSG15})$$

Then

$$-2v \log C_E \leq \log \det J_{e,s}^\chi - \log \det H_{e,s}^\chi \leq 0, \quad \log C_E = O_{\text{actual}}(q). \quad (\text{PSG16})$$

The inverse of evaluation on L_v is the Lagrange map with columns $l_b(y) = \prod_{t \neq b} (y - z_t)/(z_b - z_t)$. In the full Gamma orthonormal basis, multiplication by y has raising coefficients $\sqrt{(n+1)(n+s/2)}$ and lowering coefficients $\sqrt{n(n-1+s/2)}$, by PSG2. Through degree $v-1$ their sum has norm at most $A_{v,s}$. Starting with the constant of squared norm M_s , multiplication by each factor gives $\sum_b \|l_b\|^2 \leq M_s (C_L^{\text{int}})^2$. Cauchy–Schwarz bounds the norm of the evaluation inverse by the square root of that number.

The recurrence PSG2 also gives the complete generating identity

$$\sum_{n \geq 0} p_n(y) t^n / n! = (1 + t^2)^{-s/4} e^{y \arctan t}.$$

In fact summing the recurrence gives $(1 + t^2) \partial_t F = (y - st/2)F$, $F(0) = 1$, whose solution is precisely this series. On $|t| = \rho$, $|\arctan t| \leq \text{arctanh } \rho$. Cauchy's coefficient formula bounds the full evaluation norm through degree ν , at the chosen v nodes, by the square root of

$$\frac{v}{M_s} (1 - \rho^2)^{-s/2} e^{2Za_\rho} \sum_{n=0}^{\nu} \frac{n!}{(s/2)_n} \rho^{-2n}. \quad (\text{PSG17})$$

Every factorial and the original mass follow from division by $h_{n,s}$. For $s \geq 1$ this ratio is at most $a_n = n!/(1/2)_n$. One has $a_1 = 2$ and $a_{n+1}/a_n = (n+1)/(n+1/2) \leq \sqrt{(n+1)/n}$, since $n(n+1) \leq (n+1/2)^2$. Thus $a_n \leq 2\sqrt{n}$ for $n \geq 1$. The value at zero is one. This proves the finite sum bound in PSG15. Multiplying the evaluation bound and the bound for its inverse on L_v cancels M_s and yields C_E .

For $l \in L_v, u \in U_e$, evaluation kills u , and therefore $\|l\| \leq C_E \|l + u\|$. The attained minimum over u gives a quotient Gram on L_v between C_E^{-2} and one times its original Gram. Factoring the full Gram of $[L_v, U_e]$ in both orders gives the exact same-frame identity

$$\frac{\det J^\chi}{\det H^\chi} = \frac{\det \text{Gram}((L_v + U_e)/U_e) \text{ in the fixed } L_v \text{ lifts}}{\det \text{Gram}(L_v)}. \quad (\text{PSG18})$$

Its v eigenvalues prove PSG16. There is no omitted frame determinant: the same combined frame occurs in both factorizations. Finally v is fixed, $Z = O(k)$, $A_{v,s} = O(k)$, $\nu = O(q)$, and the nonzero Lagrange denominators are fixed. The logarithm of the explicit PSG15 expression is therefore $O(q)$.

4.5 The original factorial band

For $0 \leq a < d$ define

$$n_a = q + a - v, \quad m_a = n_a + (s-1)/2, \quad \lambda_a = \mu_v \binom{q+a}{v}, \quad \mathfrak{F}_{k,e,s} = \prod_{a=0}^{d-1} |\lambda_a|^2 h_{n_a,s}. \quad (\text{PSG19})$$

These are the exact PDS relation leading coefficients and degrees. No original relation polynomial has been replaced by its leading term. Let

$$\begin{aligned} E_{k,e,s}^{\text{band}} &= 2d\ell_s \max(|\log(2q/\theta)|, |\log(2Q_s/\theta)|) \\ &\quad + \sum_{a<d} (|n_a - q| + |m_a - q|) \log \max(q, n_a, m_a) \\ &\quad + \sum_{a<d} (3 + \log n_a + \log(2m_a) + \log m_a + \tfrac{1}{2} |\log \pi|) \\ &\quad + \tfrac{d}{2} \log 2 + 2 \sum_{a<d} |\log |\lambda_a||. \end{aligned} \quad (\text{PSG20})$$

Then PSG8 and PSG16 imply the completely finite bounds

$$\begin{aligned} -B_{k,e,s}^- - 2v \log C_E - E_{k,e,s}^{\text{band}} &\leq \log \frac{\det J_{e,s}^\chi}{\mathfrak{F}_{k,e,s}} - 2dq \log(4/\pi) \\ &\leq B_{k,e,s}^+ + E_{k,e,s}^{\text{band}}. \end{aligned} \quad (\text{PSG21})$$

To check all constants, use $(s/2)_{n_a} = \Gamma(m_a + 1/2)/\Gamma(s/2)$. The exact difference of the center in PSG8 and $\log \mathfrak{F}$ is

$$df_\theta(Q_s) - \sum_{a<d} \{\log(n_a!) + \log \Gamma(m_a + 1/2)\} - \tfrac{d}{2} \log 2 - 2 \sum_{a<d} \log |\lambda_a|. \quad (\text{PSG22})$$

The complete $d \log M_s$ and $d \log \Gamma(s/2)$ cancel here against the retained $d \log \beta_s$, by its exact PSG4 value.

For $F(x) = x \log x - x$, PSG11 gives $|\log(n!) - F(n)| \leq 1 + \log n$. The exact identity $\Gamma(m + 1/2) = \sqrt{\pi}(2m)!/(4^m m!)$ gives, by the same bounds,

$$|\log \Gamma(m + 1/2) - F(m)| \leq 2 + \log(2m) + \log m + \tfrac{1}{2} |\log \pi|.$$

All $n_a, m_a \geq 1$. Integrating $F'(x) = \log x$ between the indicated positive integer endpoints bounds their shifts from q by the second line of PSG20. Integrating $f'_\theta(x) = 2 \log(2x/\theta)$ gives its first line. Finally $f_\theta(q) - 2F(q) = 2q \log(2/\theta)$. The remaining two lines retain all factorial errors and original leading scalars. This proves PSG21.

Since $d = 8k + 24 + e$, $\ell_s = O(k)$, $n_a - q, m_a - q = O(k)$, and $|\log |\lambda_a|| = O_{\text{actual}}(\log q)$ with fixed v, μ_v , PSG20 is $O_{\text{actual}}(q \log q)$. Combining the finite bounds gives, in both original source orders and at both primitive cutoffs,

$$\boxed{\frac{1}{kq} \log \frac{\det J_{e,s}^\chi}{\mathfrak{F}_{k,e,s}} \longrightarrow 16 \log(4/\pi).} \quad (\text{PSG23})$$

The same limit holds with H^χ instead of J^χ . This evaluates a specified full original source quotient. Passing it through $\mathcal{T}_A$ requires its actual distortion on this d -dimensional source. No bound on that distortion or value for a later arithmetic allocation is assumed in PSG23.

5 The original finite Gamma translation on every small-rank plane

Fix the original simple quartet, its actual five-orbit period and logarithm branch. Keep every coefficient of the original nonreduced conductor:

$$\begin{aligned} 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad \beta_* = \frac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \\ E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{s,a,b} z}, \quad \mu_m = E_A^{(m)}(0), \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad \mu_v \neq 0, \\ k = 4l + 1 \geq 9, \quad s \in \{1, k\}, \quad c_0 = k/2, \quad c_1 = k/2 - 4. \end{aligned} \quad (\text{SGT1})$$

The coefficients, symbol, order and first nonzero moment are fixed as k varies. No particular coefficient is assumed nonzero except through the stated exact order. Every result below is for this original operator and these two original source orders. An arbitrary integer $N \geq 0$ is allowed in the finite statements.

5.1 Full Gamma measure, recurrence and coefficient isometry

Write $b = s/2 > 0$ and retain

$$\begin{aligned} M_s &= (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\ \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+b)p_{n-1,s}. \end{aligned} \quad (\text{SGT2})$$

These are exactly the measure and recurrence in PDS6 and DNE7. For completeness their generating and norm identities are derived here. The recurrence, multiplied by $t^n/n!$ and summed, gives $(1+t^2)\partial_t F = (y-bt)F$, $F(y,0) = 1$. The unique formal solution, also holomorphic for $|t| < 1$ at fixed y , is

$$F(y,t) = \sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1+t^2)^{-b/2} \exp(y \arctan t). \quad (\text{SGT3})$$

Both branches take their value zero or one at $t = 0$ as appropriate.

Here is a direct check of the full mass and norms. With $a = s/4$, the Euler gamma integrals and the change of variables $r = u + v_0$, $x = \log(u/v_0)$ in their product give

$$|\Gamma(a + i\xi)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{i\xi x} (2 \cosh(x/2))^{-2a} dx.$$

The integrand as a function of x is analytic for $|\Im x| < \pi$. Shifting the integration line by any fixed amount of modulus less than π is justified by its exponential decay at both real ends. It bounds this Fourier transform by a constant times $e^{-h|\xi|}$ for each $0 < h < \pi$. Fourier inversion is therefore absolutely convergent; it may also be obtained by inserting a Gaussian factor and passing to the limit by dominated convergence. Substituting $y = 2\xi$ and the exact multiplier in SGT2 gives

$$\int_{\mathbb{R}} e^{iuy} dm_s(y) = M_s (\cosh u)^{-b}, \quad \int_{\mathbb{R}} e^{uy} dm_s(y) = M_s (\cos u)^{-b} \quad (|u| < \pi/2). \quad (\text{SGT4})$$

For the second equality, the preceding exponential bounds give analytic continuation to the indicated strip; equality on the real Fourier axis then proves it there. In particular the mass is exactly M_s . For sufficiently small real t, z , SGT3–4 and $\cos(\arctan t + \arctan z) = (1-tz)/\sqrt{(1+t^2)(1+z^2)}$ imply

$$\int_{\mathbb{R}} F(y,t) F(y,z) dm_s(y) = M_s (1-tz)^{-b}.$$

The exponential moment bound justifies differentiating under the integral in a neighborhood of $(0, 0)$. Comparing coefficients proves

$$\begin{aligned} \int_{\mathbb{R}} p_{n,s}(y) p_{m,s}(y) dm_s(y) &= \mathbf{1}_{n=m} h_{n,s}, & h_{n,s} &= M_s n! (b)_n, \\ \phi_{n,s,c}(S) &= \frac{p_{n,s}((S-c)/i)}{\sqrt{h_{n,s}}}, & \rho_n &= \sqrt{\frac{n!}{(b)_n}} > 0. \end{aligned} \quad (\text{SGT5})$$

The $p_{n,s}$ are real monic polynomials by their recurrence. Thus these ϕ_n are orthonormal for the complex Hilbert norm in SGT2 and span each indicated polynomial degree range. Their leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$, not $1/\sqrt{h_{n,s}}$.

5.2 The original quotient map and its exact Toeplitz matrix

Let $\mathcal{P}_m = \mathbb{C}[S]_{\leq m}$, with $\mathcal{P}_{-1} = 0$. The original translation is

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad \mathcal{T}_A S^m = \sum_{j=0}^m \binom{m}{j} \mu_j S'^{m-j}. \quad (\text{SGT6})$$

The second equality is the binomial theorem followed by the exact definition of every moment. If $m \geq v$ and P has leading coefficient $p_m \neq 0$, then $\mathcal{T}_A P$ has degree $m - v$ and leading coefficient $p_m \binom{m}{v} \mu_v \neq 0$. If $m < v$ its image is zero. Consequently the following map is an isomorphism, with an onto representative map and the displayed full kernel:

$$\begin{aligned} \ker(\mathcal{T}_A : \mathcal{P}_{N+v} \longrightarrow \mathcal{P}_N) &= \mathcal{P}_{v-1}, \\ A_{N,s} : (\mathcal{P}_{N+v}/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\xrightarrow{\cong} (\mathbb{C}[S']_{\leq N}, \|\cdot\|_{s,c_1}), \\ \|[P]\|_{s,c_0,\text{quot}}^2 &= \min_{L \in \mathcal{P}_{v-1}} \|P + L\|_{s,c_0}^2. \end{aligned} \quad (\text{SGT7})$$

Surjectivity follows successively by degree from the nonzero diagonal coefficients in SGT6. The orthogonal representative of this quotient is precisely the span of $\phi_{v,s,c_0}, \dots, \phi_{N+v,s,c_0}$. It is not a quotient by an omitted arithmetic relation.

On the entire open unit disk put

$$w(t) = -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v! \neq 0. \quad (\text{SGT8})$$

The value at zero is removable by the exact order v . For $|t| < 1$, $|w(t)| \leq \arctanh |t|$ by the arctangent series. This proves holomorphy of the specified g . Applying the finite sum SGT6 to SGT3, with its polynomial argument, gives exactly

$$\begin{aligned} \sum_{m \geq 0} (\mathcal{T}_A p_{m,s}((S - c_0)/i))(S') \frac{t^m}{m!} &= (1 + t^2)^{-b/2} e^{(S' - c_0)w(t)} E_A(w(t)) \\ &= t^v g(t) (1 + t^2)^{-b/2} e^{(S' - c_1)w(t)}. \end{aligned} \quad (\text{SGT9})$$

The center factor is e^{-4w} because $c_0 - c_1 = 4$. No inverse of $\mathcal{T}_A$, or scalar α_A , has been inserted in this identity.

For a holomorphic germ $u(t) = \sum_{j \geq 0} u_j t^j$ define $T_N(u)_{r,n} = u_{n-r}$ for $0 \leq r \leq n \leq N$, and zero for $r > n$. Coefficients of SGT9 and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give the complete matrix of SGT7 in its two orthonormal frames:

$$\begin{aligned} A_{N,s} &= D_{\text{out}}^{-1} T_N(g) D_{\text{in}}, \\ D_{\text{out}} &= \text{diag}(\rho_0, \dots, \rho_N), \quad D_{\text{in}} = \text{diag}(\rho_v, \dots, \rho_{N+v}), \\ (A_{N,s})_{r,n} &= \begin{cases} g_{n-r} \rho_{n+v} / \rho_r, & r \leq n, \\ 0, & r > n. \end{cases} \end{aligned} \quad (\text{SGT10})$$

Each source and target frame contains $1/\sqrt{M_s}$. Those identical factors cancel only in this coefficient matrix, since s is unchanged. All actual squared norms still use the full SGT2 measure. In particular, for $N = 0$ the matrix is the literal scalar $(-i)^v \mu_v \rho_v / v!$.

5.3 A proved original-data zero-free disk

The following constants use every original coefficient and shift:

$$\begin{aligned} A_{\text{abs}} &= \sum_{a,b} |a_{ab}|, & B &= \max\{1, \max_{a,b} |\beta_{8,a,b}|\}, & C_v &= \frac{A_{\text{abs}} B^v}{|\mu_v|} \geq 1, \\ R_0 &= \frac{1}{B} \min\{1, (v+1)/(2 \exp(1) C_v)\}, & a_0 &= \min\{1/2, \tanh(R_0/2)\}, \\ d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)} \geq 8/9, & M_0 &= \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}. \end{aligned} \quad (\text{SGT11})$$

All are finite, R_0, a_0, d_0, M_0 are positive, and $a_0 < 1$. Indeed $|\mu_v| \leq A_{\text{abs}} B^v$ proves the stated bound on C_v . Define the entire removable quotient $B_\infty(z) = v! E_A(z)/(\mu_v z^v)$, with value one at zero. Its full Taylor remainder is bounded by

$$|B_\infty(z) - 1| \leq \frac{C_v B |z|}{v+1} e^{B|z|} \leq \frac{1}{2} \quad (|z| \leq R_0). \quad (\text{SGT12})$$

For the first inequality expand each exponential, use $|\mu_m| \leq A_{\text{abs}} B^m$ and $v!/(v+1+j)! \leq 1/((v+1)j!)$, and sum the exponential series. The two inequalities $BR_0 \leq 1$ and $C_v BR_0/(v+1) \leq 1/(2 \exp(1))$ prove the second. This is the complete RDS11 disk proof for the unchanged symbol.

On $|t| \leq a_0$ we have $|w(t)| \leq R_0/2$ and, also at zero,

$$\left| \frac{\arctan t}{t} - 1 \right| \leq \sum_{j \geq 1} \frac{a_0^{2j}}{2j+1} \leq \frac{a_0^2}{3(1-a_0^2)}.$$

Thus $|w(t)/t| \geq d_0$. Factoring SGT8 as $g(t) = (\mu_v/v!)(w(t)/t)^v e^{-4w(t)} B_\infty(w(t))$ shows that it has no zero on that entire closed disk and proves

$$|g(t)| \geq \frac{|\mu_v|}{2v!} d_0^v e^{-2R_0}, \quad |1/g(t)| \leq M_0 \quad (|t| \leq a_0). \quad (\text{SGT13})$$

This supplies the zero-free disk and the inverse bound from the actual finite symbol rather than assuming them.

5.4 Every actual pole and its finite-rank Toeplitz contribution

Call $0 < a < 1$ admissible when g has no zero on $|t| = a$. Zeros of a nonzero holomorphic function are isolated; therefore there are only finitely many in each compact subdisk. There are admissible radii arbitrarily close to one. For a fixed such a , let $\mathcal{Z}_a$ be the exact zeros ζ of g with $|\zeta| < a$, and let m_ζ be their orders. Zero is not among them. Retain every multiplicity and define the actual Laurent coefficients

$$\begin{aligned} H_a &= \sum_{\zeta \in \mathcal{Z}_a} m_\zeta, & A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \left(\frac{(t - \zeta)^{m_\zeta}}{g(t)} \right) \Big|_{t=\zeta}, & 1 \leq h \leq m_\zeta, \\ f &= 1/g, & f_a(t) &= f(t) - \sum_{\zeta \in \mathcal{Z}_a} \sum_{h=1}^{m_\zeta} \frac{A_{\zeta,h}}{(t - \zeta)^h}. \end{aligned} \quad (\text{SGT14})$$

The factors t^{-v} and e^{-4w} in g create no nonzero zero. Moreover $w'(t) = -i/(1+t^2) \neq 0$ and $e^{-2w(t)} = (1+it)/(1-it)$ proves injectivity of w on the disk. Thus these poles are precisely the symbol zeros already identified in PEC3-5, under this exact change of variable, with their same orders. The extra center factor is fully retained in every $A_{\zeta,h}$. The function f_a is holomorphic on a neighborhood of the closed a disk, since all its interior principal parts have been subtracted and no boundary zero exists.

Here are completely specified finite constants on that circle:

$$\begin{aligned}
W_a &= \operatorname{arctanh} a, \quad m_{E,a} = \min_{|t|=a} |E_A(w(t))| > 0, \quad G_a = A_{\text{abs}} a^{-v} e^{(B+4)W_a}, \\
M_a &= \frac{a^v e^{4W_a}}{m_{E,a}} + \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(a - |\zeta|)^h}, \quad P_a = \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(1 - |\zeta|)^h}, \\
S_N(a) &= \sum_{j=0}^N a^{-j} = a^{-N} \frac{1 - a^{N+1}}{1 - a}.
\end{aligned} \tag{SGT15}$$

Empty pole sums are zero. The minimum is the same actual finite-circle quantity used in PEC4, and is positive by continuity and admissibility. It is not a new uniform zero-free assertion. The triangle inequality on that circle gives $|g| \leq G_a$ and $|f_a| \leq M_a$. Every constant in SGT15 is defined by the original fixed function and its exact finitely many interior poles; none depends on k, N or s .

For one term $A/(t - \zeta)^h$, its Taylor coefficients at zero are

$$c_n = A(-1)^h \zeta^{-(n+h)} \binom{n+h-1}{h-1} \quad (n \geq 0). \tag{SGT16}$$

Extend the binomial as the polynomial $\binom{x+h-1}{h-1} = \prod_{\ell=1}^{h-1} (x+\ell)/(h-1)!$ to every integer x , also for $h=1$ by the empty product. The full matrix $C_{i,j} = c_{j-i}$, $0 \leq i, j \leq N$, has rank at most h : after multiplying rows by ζ^{-i} and columns by ζ^j , its entries are a polynomial of total degree at most $h-1$ in $j-i$. Expanding that polynomial in powers of i gives at most h column vectors, independently of N . Equivalently its column space is contained in $\operatorname{span}\{(\zeta^i i^d)_i : 0 \leq d < h\}$. For all terms belonging to one pole, these nested spaces give combined rank at most m_ζ , rather than the sum of the integers h .

For a lower-diagonal gap $\ell = i - j \geq 1$, direct substitution in the polynomial binomial gives

$$c_{-\ell} = \begin{cases} 0, & 1 \leq \ell < h, \\ -A\zeta^{\ell-h} \binom{\ell-1}{h-1}, & \ell \geq h. \end{cases} \tag{SGT17}$$

For $\ell \geq h$ each factor in the binomial is negative, and their $(-1)^{h-1}$ sign combines with $(-1)^h$ to give the displayed minus sign. For $1 \leq \ell < h$ one factor is zero. Consequently the upper triangular Taylor matrix of this pole is $C + R$, where R is strictly lower triangular and its gap- ℓ coefficient is $A\zeta^{\ell-h} \binom{\ell-1}{h-1}$ for $\ell \geq h$. If J is the upper shift with $J_{i,i+1} = 1$, then $(J^*)^\ell$ has norm at most one. The absolutely convergent binomial series therefore proves

$$\|R\| \leq |A| \sum_{\ell=h}^{\infty} |\zeta|^{\ell-h} \binom{\ell-1}{h-1} = \frac{|A|}{(1 - |\zeta|)^h}. \tag{SGT18}$$

The series identity follows by differentiating the geometric series $h-1$ times, or multiplying h copies of it and collecting coefficients. All powers and signs are those of the original pole; no principal part has been deleted from the actual Toeplitz inverse.

Since $T_N(u) = \sum_{j=0}^N u_j J^j$, multiplication of these finite matrices is truncated power-series multiplication. In particular $T_N(g)^{-1} = T_N(f)$, because $g(0) \neq 0$ and $gf = 1$ as germs. Cauchy's coefficient estimate, SGT13 and SGT15–18 prove

$$\begin{aligned}
\|T_N(g)\| &\leq F_{a,N} := G_a S_N(a), \\
\|T_N(f)\| &\leq G_{0,N} := M_0 S_N(a_0), \\
T_N(f) &= C_{a,N} + R_{a,N}, \quad \operatorname{rank} C_{a,N} \leq H_a, \quad \|R_{a,N}\| \leq V_{a,N} := M_a S_N(a) + P_a.
\end{aligned} \tag{SGT19}$$

Indeed sum the coefficient estimates times $\|J^j\| \leq 1$ for the analytic terms, and sum the exact pole ranks and lower-triangular remainders just proved. The total inverse matrix is unchanged.

5.5 Finite exterior bounds, including the expensive directions

Write $d = N + 1$ and arrange singular values in decreasing order. If a matrix C has rank at most H , then $\ker C$ has codimension at most H . For $x \in \ker C$, $\|(C + R)x\| \leq \|R\|\|x\|$. The variational description of singular values therefore gives $\sigma_j(C + R) \leq \|R\|$ for $j > H$. To verify this description directly, diagonalize $(C + R)^*(C + R)$: the complement of its first $j - 1$ eigenvectors attains the upper bound, and every subspace of codimension $j - 1$ intersects the first j eigenvectors for the reverse inequality. Applying this argument to SGT19 and retaining the separate global bound gives

$$h = \min\{r, H_a\}, \quad 0 \leq r \leq d, \quad V_{a,N}^{\text{cap}} = \min\{G_{0,N}, V_{a,N}\},$$

$$\left\| \bigwedge^r T_N(g) \right\| \leq F_{a,N}^r, \quad \left\| \bigwedge^r T_N(f) \right\| \leq G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}. \quad (\text{SGT20})$$

The norm of the r th exterior power is the product of the r largest singular values: this follows by applying a singular-value decomposition to the orthonormal wedge bases. Only at most H_a directions use the zero-free-radius bound. If $H_a > d$, then $h = r$ and there is no tail factor. Every constant is positive; the empty exterior power has norm one.

Retain the exact diagonal constants from the original norms:

$$\alpha_+ = \left(\max_{0 \leq n \leq N} \rho_{n+v} \right) \left(\max_{0 \leq n \leq N} \rho_n^{-1} \right), \quad \alpha_- = \left(\max_{0 \leq n \leq N} \rho_{n+v}^{-1} \right) \left(\max_{0 \leq n \leq N} \rho_n \right),$$

$$L_{s,N,v} = \frac{1}{2} \max\{1, s/2 - 1\} \mathfrak{H}_{N+v}, \quad \mathfrak{H}_m = \sum_{j=1}^m 1/j, \quad \mathfrak{H}_0 = 0. \quad (\text{SGT21})$$

For $s = k \geq 9$, the product defining ρ_n and $\log(1 + x) \leq x$ imply $|\log \rho_n| \leq (b - 1)\mathfrak{H}_n/2$. For $s = 1$, its logarithm is positive and $\log((j + 1)/(j + 1/2)) \leq 1/(2j + 1) \leq 1/(j + 1)$ gives $|\log \rho_n| \leq \mathfrak{H}_n/2$. Thus $|\log \rho_n| \leq L_{s,N,v}$ for all $0 \leq n \leq N + v$, including $n = 0$. In particular $|\log \alpha_{\pm}| \leq 2L_{s,N,v}$. Applying the two diagonal maps in SGT10 on exterior powers gives

$$\left\| \bigwedge^r A_{N,s} \right\| \leq \alpha_+^r F_{a,N}^r,$$

$$\left\| \bigwedge^r A_{N,s}^{-1} \right\| \leq \alpha_-^r G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}. \quad (\text{SGT22})$$

This retains the rectangular degree shift $n \mapsto n + v$ exactly.

Let $[P_1], \dots, [P_r]$ be any independent original classes in SGT7 and use that same ordered frame in both Grams:

$$H_{\text{in}} = (\langle [P_i], [P_j] \rangle_{s, c_0, \text{quot}})_{i,j}, \quad H_{\text{out}} = (\langle \mathcal{T}_A P_i, \mathcal{T}_A P_j \rangle_{s, c_1})_{i,j}, \quad \Delta_{N,s}(\mathbf{P}) = \log \frac{\det H_{\text{out}}}{\det H_{\text{in}}}. \quad (\text{SGT23})$$

The inner products are conjugate-linear in their first variable. Both Grams are positive, by independence and SGT7. Exterior volumes and their inverse give the finite, potentially asymmetric bounds

$$-2r \log \alpha_- - 2h \log G_{0,N} - 2(r - h) \log V_{a,N}^{\text{cap}} \leq \Delta_{N,s}(\mathbf{P})$$

$$\leq 2r \log \alpha_+ + 2r \log F_{a,N}. \quad (\text{SGT24})$$

For proof, express the classes in their SGT7 orthonormal quotient frame as columns X . Their squared wedge norm is $\det(X^* X)$, and the squared norm of their image wedge is $\det(X^* A_{N,s}^* A_{N,s} X)$. Apply SGT22 to that wedge and then to its image with $A_{N,s}^{-1}$. A change of the original ordered frame by a matrix U multiplies both determinants by $|\det U|^2$; the ratio therefore retains the original frame and is frame invariant. No specially conditioned plane has been selected.

For a nonnegative enclosure write $\log^+ x = \max\{0, \log x\}$ and set

$$\mathcal{U}_{a,N,s,r} = 4r L_{s,N,v} + 2r \log^+ F_{a,N},$$

$$\mathcal{L}_{a,N,s,r} = 4r L_{s,N,v} + 2h \log^+ G_{0,N} + 2(r - h) \log^+ V_{a,N}^{\text{cap}}. \quad (\text{SGT25})$$

Then $-\mathcal{L} \leq \Delta_{N,s}(\mathbf{P}) \leq \mathcal{U}$. These are finite explicit constants even for $r = 0$, with value zero for that empty Gram ratio. All maxima, minima, poles and residues in the constants refer to the fixed original symbol SGT1.

5.6 Uniform small-rank volume distortion and the actual PDS frame

Fix finite constants $C_N, C_r > 0$. Uniformly over $0 \leq N \leq C_N k^2$, $0 \leq r \leq \min\{C_r k, N+1\}$, $s = 1$ or k , and every original rank- r plane in SGT7,

$$\sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \rightarrow 0 \quad (k = 4l+1 \rightarrow \infty). \quad (\text{SGT26})$$

Here the supremum is over precisely those finite-degree and rank domains, including all coefficient choices for the planes. We give both limiting estimates. For each fixed admissible a , $S_N(a) \leq (N+1)a^{-N}$ and $a^{-N} \geq 1$, so

$$\begin{aligned} \log^+ F_{a,N} &\leq N \log(1/a) + \log(N+1) + \log^+ G_a, \\ \log^+ V_{a,N}^{\text{cap}} &\leq \log^+ V_{a,N} \leq N \log(1/a) + \log(N+1) + \log^+(M_a + P_a), \\ \log^+ G_{0,N} &\leq N \log(1/a_0) + \log(N+1) + \log^+ M_0. \end{aligned} \quad (\text{SGT27})$$

Also $\mathfrak{H}_m \leq 1 + \log(m+1)$, by integral comparison. The common diagonal error $4rL$ is therefore $O_{C_N, C_r, v}(k^2 \log(k+1))$. The global inverse term has $h \leq H_a$, which is fixed before the k limit, so it is $O_{\text{actual}, a, C_N}(k^2 + \log(k+1))$. It follows from SGT25 and SGT27 that

$$\limsup_{k \rightarrow \infty} \sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \leq 2C_r C_N \log(1/a). \quad (\text{SGT28})$$

The constants in this step may depend on this fixed a . Only after this limit do we take admissible radii arbitrarily close to one, as proved before SGT14. The right side then tends to zero. This proves SGT26 without assuming a zero-free unit disk or a uniform pole count as a tends to one.

For the original PDS quotient set

$$\begin{aligned} q_k &= (k+1)^2, \quad Q = (k+5)^2, \quad e \in \{0, 1\}, \\ N_e &= Q - v - 1 + e, \quad \ell_e = 8k + 23 + e, \quad d_e = \ell_e + 1 = 8k + 24 + e, \\ P_a(S) &= \chi_k(S) S^a, \quad r_a(S') = \mathcal{T}_A P_a(S'), \quad 0 \leq a \leq \ell_e, \\ \chi_k(S) &= \prod_{u,w=0}^k (S - \beta_{k,u,w}). \end{aligned} \quad (\text{SGT29})$$

Then $q_k + \ell_e = N_e + v = Q - 1 + e$. A nonzero linear combination of the P_a has degree at least $q_k > v - 1$ because it is χ_k times a nonzero polynomial. Thus their classes modulo $\mathcal{P}_{v-1}$ are independent, of exactly the original rank d_e , without a minor or genericity assumption.

Let X_e be their full column matrix in the original c_0 orthonormal frame of degrees $0, \dots, N_e + v$, and write $P_{<v}$ and $P_{\geq v}$ for the row selections in that frame. The exact source and image Grams are

$$\begin{aligned} Y_e &= P_{\geq v} X_e, \quad H_{\text{src},e} = Y_e^* Y_e = X_e^* X_e - (P_{<v} X_e)^* (P_{<v} X_e) > 0, \\ C_e &= A_{N_e,s} Y_e, \quad H_e = C_e^* C_e, \quad -\mathcal{L}_{a,N_e,s,d_e} \leq \log \frac{\det H_e}{\det H_{\text{src},e}} \leq \mathcal{U}_{a,N_e,s,d_e}. \end{aligned} \quad (\text{SGT30})$$

The first equality is the attained full quotient minimum over the entire original $\mathcal{P}_{v-1}$. The next equality is the coefficient map SGT10, since those removed low columns are killed by $\mathcal{T}_A$. The resulting C_e, H_e are exactly PDS7, including the target center c_1 and both original orders. For $k \geq 9$, $Q \leq (14k/9)^2 < 3k^2$, so $N_e \leq 3k^2$ and $d_e \leq 11k$. Hence SGT26 applies directly:

$$\log \frac{\det H_e}{\det H_{\text{src},e}} = o(k^3) = o(kq_k) \quad (e = 0, 1, s = 1, k), \quad (\text{SGT31})$$

uniformly over those four original source/endpoint choices. The full original mass has not been removed from either Gram.

The precise dual statement also retains all PDS conjugations: for the complex-linear functional $x^\top u$ on a Gram H , its squared dual norm is $x^* \overline{H}^{-1} x$. Thus the same coefficient coordinates carry

$$G_e = \overline{H_e^{-1}}, \quad G_{\text{src},e} = \overline{H_{\text{src},e}^{-1}}, \quad \log \frac{\det G_e}{\det G_{\text{src},e}} = -\log \frac{\det H_e}{\det H_{\text{src},e}}. \quad (\text{SGT32})$$

This is an equality of the original full quotient-source dual metrics, not a statement about a later connecting-target Gram.

For the endpoint $e = 1$ and the original four PDS8 rows, retain the same full rank-four matrix W and choose any basis Z of $\ker W^\top$. The polynomial quotient by those four columns, in the dual coordinates $a = Z^\top f$, has metric $(Z^\top H^{-1} \overline{Z})^{-1}$ by minimizing $f^* H f$. Consequently it is the conjugate inverse of the restriction $Z^* \overline{H}^{-1} Z$, exactly as in PDS9. For the paired forms $H_e, H_{\text{src},e}$, fixed complementary frames or the same attained Schur decomposition therefore give

$$\begin{aligned} \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} &= -\Delta_{N_e,s}(\mathbf{P}) + \log \frac{\det(W^* H_e W)}{\det(W^* H_{\text{src},e} W)}, \\ -\mathcal{U}_{a,N_e,s,d_e} - \mathcal{L}_{a,N_e,s,4} &\leq \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} \leq \mathcal{L}_{a,N_e,s,d_e} + \mathcal{U}_{a,N_e,s,4}. \end{aligned} \quad (\text{SGT33})$$

Indeed $\ker Z^\top = \text{im } W$. If J is any fixed right inverse of $Z^\top$, then $[W, J]$ is invertible, and block Schur factorization gives $\det((Z^\top H^{-1} \overline{Z})^{-1}) = |\det[W, J]|^2 \det H / \det(W^* H W)$. Taking the ratio cancels this exact frame determinant and proves the first equality. The W columns correspond to the four independent original polynomials $\chi_k R_i$ modulo $\mathcal{P}_{v-1}$; SGT24–25 applied to that actual rank-four plane prove the interval. At $e = 1$ this is the original primitive source, and its determinant distortion is again $o(kq_k)$. At $e = 0$ the primitive condition is empty and SGT32 already gives its full statement.

The earlier PEC1–18 theorem uses actual numerator-jet cancellation at symbol zeros to control a positive dual spectrum. Here SGT7–10 first proves the polynomial source/target morphism, and SGT16–22 retains all principal parts in an exact finite-rank Toeplitz summand, giving both directions on every small-rank plane. Neither statement replaces the original lower quotient, the four primitive rows, or the connecting-target metric. The original-data zero-free disk is the RDS11 disk with its full proof repeated in SGT11–13.

6 The surviving boundary quotient and its original arithmetic kernel

Retain the actual simple quartet, its proved five-orbit period and logarithm branch, every coefficient of the original nonreduced conductor, and the complete CIT, CNC, DFX, KRC, DNE, NBG and PCJ maps. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad m_a = 16a - 48 - v, \\ 0 &\leq v = \text{ord}_0 E_A \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \\ (s_0, s_1) &= (1, 1) \text{ or } (k, j), \quad c'_a = a/2 - 4, \quad D \geq q_j - v - 1. \end{aligned} \quad (\text{RQB1})$$

All coefficient spaces and dual functionals are complex-linear. No star replaces an ordinary transpose in an original algebraic map. Metric stars are complex adjoints. The degree progression and the common cutoff D are literal, including the two distinct source measures. Write $s_k = s_0$ and $s_j = s_1$ when source orders have degree subscripts.

6.1 The exact coefficient kernel, rank and original kernel quotient

Let $V = \mathbb{C}^{m_k}$, $U = \mathbb{C}^{4m_j}$ and retain

$$\begin{aligned} B_0 &= \mathbf{B}_k, \quad B_1 = \mathbf{B}_j, \quad J_0 = \mathbf{J}_k, \quad J_1 = \mathbf{J}_j, \quad \ker B_a = \text{im } J_a \quad (a = 0, 1), \\ M_i &= (\alpha_A N_j Z_{\text{exp},j})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp},k}), \quad m_{i,k}[f] = [x_i^4 f], \\ B_{\text{cof}} &= \text{stack}_{i=1}^4 M_i, \quad \mathbb{B} = \text{diag}(B_1, 4), \quad X_{\mathbb{J}} = \text{diag}(J_1, 4), \quad \mathbb{J} = \text{im } X_{\mathbb{J}}, \\ \Phi &= \mathbb{B} B_{\text{cof}}, \quad J = \text{im } J_0, \quad \mathcal{L} = \ker \Phi, \quad Y = J + \mathcal{L}, \quad E = \ker(\Phi J_0). \end{aligned} \quad (\text{RQB2})$$

Here $\text{diag}(P, 4)$ means four copies. The original KRC bijections specify the inverses onto their full images; both occurrences of α_A are retained before their proved cancellation in M_i . CNC6 or DFX21 proves B_{cof} injective on the entire V . All $\mathbf{B}_a$ are onto their actual kernel-dual coordinate spaces. Write $t_{K,a}$ for their ranks and $j_{0,a}$ for the column ranks of $\mathbf{J}_a$; the original exact values are $t_{K,a} = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$.

Set exactly as in CIT24

$$W = \mathbb{J} + \text{im } B_{\text{cof}}, \quad W_{\text{inv}} = \mathbb{J} + \text{im}(B_{\text{cof}}J_0).$$

The literal map and its full kernel are

$$\begin{aligned} \pi_b : V &\longrightarrow W/W_{\text{inv}}, & x &\longmapsto [B_{\text{cof}}x], & \ker \pi_b &= Y, \\ V/Y &\xrightarrow{\sim} W/W_{\text{inv}} \xrightarrow{\sim} \text{im } \Phi / \text{im}(\Phi J_0), & [x] &\longmapsto [B_{\text{cof}}x] \longmapsto [\Phi x]. \end{aligned} \quad (\text{RQB3})$$

Indeed $B_{\text{cof}}x \in W_{\text{inv}}$ precisely when $B_{\text{cof}}(x - J_0a) \in \mathbb{J}$ for some a , equivalently $x - J_0a \in \ker \Phi$. This proves the first kernel and both surjections. The second arrow has kernel exactly $\mathbb{J}$ before the displayed quotient, because $\ker \mathbb{B} = \mathbb{J}$. Thus it is injective on the stated quotient.

Put $\ell = \dim \mathcal{L}$, $\epsilon = \dim E$ and $r_b = \dim(W/W_{\text{inv}})$. Since J_0 is injective, it identifies E with $J \cap \mathcal{L}$. Consequently

$$\begin{aligned} r_b &= m_k - j_{0,k} - \ell + \epsilon = t_{K,k} - \ell + \epsilon = \text{rank } \Phi - \text{rank}(\Phi J_0), \\ 0 &\longrightarrow B_0\mathcal{L} \longrightarrow \mathbb{C}^{t_{K,k}} \xrightarrow{[B_0x] \mapsto [B_{\text{cof}}x]} W/W_{\text{inv}} \longrightarrow 0, & \dim B_0\mathcal{L} &= \ell - \epsilon. \end{aligned} \quad (\text{RQB4})$$

The middle arrow is well defined because changing x by $\ker B_0 = J$ changes its image by W_{inv} . Its kernel is $B_0Y = B_0\mathcal{L}$. Under KRC4–5 this is the exact quotient of $(K_k/(K_k \cap L_k))^\vee$, where $L_k = \mathbb{C}[S]_{<v}$ is the original polynomial low space. Thus the original low overlap $t_{0,k}$ remains in both the domain rank and its actual quotient.

Here is a finite calculation of $\mathcal{L}$ using the input polynomials. Use the exact Segre coefficient map $\Psi_a : S_{x,a} \rightarrow T_{a,a}$ of DFX13, the selector U_a^{inv} of degree- a monomials with $\sum i\nu_i \equiv 0 \pmod{5}$, and the complete original primary conductor convolution $C_a^\top$. Let $\text{quo}(G)$ be the following exact row-pivot quotient: take the first independent columns F of G , the first row set $\mathcal{R}$ with $F_{\mathcal{R}}$ invertible, and send x to $(x - FF_{\mathcal{R}}^{-1}x_{\mathcal{R}})_{\mathcal{R}^c}$. It is onto and has kernel $\text{im } G$, since the subtracted vector agrees with x on $\mathcal{R}$ and a zero remainder means x is that same column combination. Empty and full ranks give the identity and zero map respectively. Define

$$\begin{aligned} P_j^{\text{inv}} &= \text{quo}[C_j^\top, \Psi_j U_j^{\text{inv}}], \\ (H_{i,k}\nu)_{ab} &= \sum_{r,t=0}^4 b_{i,r,t} \nu_{a-r,b-t}, & b_{i,r,t} &= [Z^r W^t] L_i(Z, W)^4, \\ \mathcal{A}_k &= \text{stack}_{i=1}^4 (P_j^{\text{inv}} H_{i,k} Z_{\text{exp},k}), & \mathcal{L} &= \ker \mathcal{A}_k, & E &= \ker(\mathcal{A}_k J_0). \end{aligned} \quad (\text{RQB5})$$

Every off-rectangle coefficient in the convolution is zero. Membership of $H_{i,k} Z_{\text{exp},k} x$ in the displayed column space means its full conductor class has an invariant representative. Its first v jets already vanish by Leibniz multiplication; the conductor columns have zero such jets, so the invariant representative has zero jets as well. It is therefore exactly membership in $\mathcal{J}_{v,j}$, which KRC5 identifies with $\text{im } J_1$. This proves RQB5 and shows that the two computed kernels are those in RQB2, including all exceptional coefficient minors. In particular the first line of RQB4 is an exact finite rank formula in the original coefficients, without a chosen generic rank.

6.2 A branch constraint for arbitrary source numerators

Work first in $S_x = \mathbb{C}[x_1, x_2, x_3, x_4]$, with the original H , $Q = Q_0(H^{-1}x)$, $Q_a = g^a Q$ and $\mathcal{A} = \prod_{a=1}^4 Q_a$, where

$$gf(x) = f(\text{diag}(\zeta^{-1}, \zeta^{-2}, \zeta^{-3}, \zeta^{-4})x).$$

The primary order in the Segre map is unchanged. In the polynomial UFD $T = \mathbb{C}[z_0, z_1, w_0, w_1]$ put

$$\psi f = f(H(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top), \quad \ell_i = \psi x_i, \quad B_a = \psi Q_a, \quad A^{\text{hom}} = \prod_{a=1}^4 B_a. \quad (\text{RQB6})$$

The restriction map is onto $T_{d,d}$ with kernel $QS_{x,d-2}$: each biform monomial has the lift in KRC13, and two lifts with the same exponents differ by the relation $t_1 t_4 - t_2 t_3$. Grouping terms with the same image proves the full kernel, as in DFX13. Hence the complete branch quotient is $T_{d,d}/B_a T_{d-2,d-2}$; no radical is taken.

The four ℓ_i are linearly independent $(1,1)$ biforms, because their coefficient matrix is the actual invertible H . Any irreducible common factor of three has bidegree $(1,0)$, $(0,1)$ or $(1,1)$: factors of a bihomogeneous polynomial are bihomogeneous, as follows by comparing least and greatest degree in each grading in a domain. Multiples of a fixed factor of these types inside $T_{1,1}$ have dimensions two, two or one. Three independent rows cannot be contained there. It follows that any irreducible factor divides at most two ℓ_i , and therefore

$$\gcd\{\ell_i^4 \ell_t^4 : 1 \leq i < t \leq 4\} = 1 \quad \text{in } T. \quad (\text{RQB7})$$

For each irreducible factor, choose two indices outside the at most two rows divisible by it. Their pair product is not divisible by it. This proves RQB7 in the biform polynomial UFD, without transferring factorization to the homogeneous quadric quotient ring.

Let $[f] \in \mathcal{C}_{v,k}$ be the image of a vector of $\mathcal{L}$. For each i there is an invariant p_i such that $x_i^4 f - p_i \in (Q, \mathcal{A})$. Fix $a \in \{1, 2, 3, 4\}$ and reduce modulo (Q, Q_a) . The ideal $(Q, \mathcal{A})$ maps to zero there; applying g^a to it also maps to zero, because $g^a Q = Q_a$ and $g^a \mathcal{A}$ contains Q as the factor with original index $5-a$. Since $g^a p_i = p_i$, the two reductions give

$$x_i^4 f = \lambda_{i,a} x_i^4 g^a f, \quad \lambda_{i,a} = \zeta^{-4ia} \mod (Q, Q_a). \quad (\text{RQB8})$$

For fixed nonzero a the four $\lambda_{i,a}$ are nonzero and pairwise distinct. Multiply the equation with index i by x_t^4 and that with index t by x_i^4 , then subtract. It follows first that $x_i^4 x_t^4 g^a f = 0$, and then that $x_i^4 x_t^4 f = 0$ in this same full branch quotient. On applying ψ , every irreducible factor P of B_a , with its actual exponent e_P , therefore satisfies

$$e_P \leq \text{ord}_P(\ell_i^4 \ell_t^4 \psi f) \quad \text{for every } i < t.$$

RQB7 supplies a pair whose multiplier has exponent zero. Thus $\text{ord}_P \psi f \geq e_P$ for every P , so B_a divides ψf with all its multiplicities. Repeating for all four a proves the new arbitrary-source constraint

$$\mathsf{T}_k \mathcal{L} \subset \mathcal{H}_{v,k} := \mathcal{C}_{v,k} \cap \ker \mathsf{j}_k, \quad \mathsf{T}_k = \alpha_A N_k Z_{\text{exp},k}. \quad (\text{RQB9})$$

DFX24 proves its invariant-source specialization. The proof above retains the arbitrary f and $g^a f$ as two different branch values until the pair equations eliminate them.

The constraint is computable on the complete nonreduced factor scheme. Let

$$\mathcal{L}_{\text{br}} = \text{lcm}(B_1, B_2, B_3, B_4), \quad (b_z, b_w) = \text{bideg } \mathcal{L}_{\text{br}}, \quad G_{\text{br}} = A^{\text{hom}} / \mathcal{L}_{\text{br}}.$$

Retain the actual scalar in A^{hom} and compensating factor units in this equation. Divisibility gives the exact kernel

$$\ker \mathsf{j}_k \simeq T_{k-b_z, k-b_w} / G_{\text{br}} T_{k-8, k-8}, \quad [u] \mapsto [\mathcal{L}_{\text{br}} u], \quad h_k = (k - b_z + 1)(k - b_w + 1) - (k - 7)^2. \quad (\text{RQB10})$$

Indeed all four branch classes vanish exactly when their biform representative is divisible by this lcm. Such representatives differ by A^{hom} exactly by the stated denominator. Nonzero biform multiplication is injective, proving the dimension.

Choose a quotient monomial basis in RQB10 by the specified row-pivot rule, multiply by $\mathcal{L}_{\text{br}}$, take its affine coefficient array and apply the original jet rows $\beta_{k,a,b}^n$, $0 \leq n < v$. Call this full matrix $\mathcal{J}_{\text{br},v}$. The denominator is killed since it is a multiple of E_A after the original exponential substitution. Thus, with $h_{v,k} = \dim \mathcal{H}_{v,k}$,

$$h_{v,k} = h_k - \text{rank } \mathcal{J}_{\text{br},v}, \quad d_{v,k} = \dim(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k}), \quad 0 \leq \ell - \epsilon \leq h_{v,k} - d_{v,k}. \quad (\text{RQB11})$$

For example $d_{v,k}$ is exactly the nullity of the four branch quotient matrices on $\mathbf{T}_k J_0$, computed with DFX30/KRC13. The last inequality follows from the injective map $\mathcal{L}/(\mathcal{L} \cap J) \rightarrow \mathcal{H}_{v,k}/(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k})$ induced by the original $\mathbf{T}_k$. Its kernel is precisely $\mathcal{L} \cap J$, so no dimension of an overlapping low space is dropped. The resulting finite rank control is

$$\max\{0, t_{K,k} - h_{v,k} + d_{v,k}\} \leq r_b \leq t_{K,k}, \quad r_b = t_{K,k} - \text{rank}(B_0 Z_{\mathcal{L}}), \quad (\text{RQB12})$$

where $Z_{\mathcal{L}}$ is the exact kernel frame from RQB5. This preserves every shared-component defect. It gives full rank exactly on the coefficient strata where $B_0 Z_{\mathcal{L}} = 0$; that criterion is a computed matrix equality, not a replacement of a nonreduced conductor by its reduced support.

6.3 Full native and raw Gamma realizations

Use the actual fixed ITL4 complement Z_R of E , so $I_R = B_{\text{cof}} J_0 Z_R$. Choose a fixed complement E_b of Y in V by exact coefficient elimination, and set

$$R_b = B_{\text{cof}} E_b, \quad Y_{\text{inv}} = [X_{\mathbb{J}}, I_R], \quad Y_W = [X_{\mathbb{J}}, I_R, R_b]. \quad (\text{RQB13})$$

These are full frames of W_{inv} and W : applying $\mathbb{B}$ kills $X_{\mathbb{J}}$, makes $\Phi J_0 Z_R$ independent, and RQB3 makes the last quotient columns independent. In particular R_b represents exactly r_b quotient values. For the full target DNE form set

$$\begin{aligned} \mathcal{D}_D &= \text{diag}(D_{j,D}^{(s_1)}, 4), \quad Q_b^E(D) = R_b^* \mathcal{D}_D R_b - R_b^* \mathcal{D}_D Y_{\text{inv}} (Y_{\text{inv}}^* \mathcal{D}_D Y_{\text{inv}})^{-1} Y_{\text{inv}}^* \mathcal{D}_D R_b, \\ \mathcal{H}_D &= \text{diag}(H_{B,j,D}, 4), \quad H_{B,a,D} = (\mathbf{B}_a(D_{a,D}^{(s_a)})^{-1} \mathbf{B}_a^*)^{-1} = \widehat{H}_{K,a,D}^{-1}, \\ U_R &= \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad Q_b^E(D) = U_b^* \mathcal{H}_D U_b - U_b^* \mathcal{H}_D U_R (U_R^* \mathcal{H}_D U_R)^{-1} U_R^* \mathcal{H}_D U_b. \end{aligned} \quad (\text{RQB14})$$

Both formulas are attained minima: the first minimizes over the complete W_{inv} ; the second first minimizes over $\ker \mathbb{B} = \mathbb{J}$, then over its remaining image U_R . The fibres are identical by RQB3, so the formulas agree. Every inverse is justified by the full frames and DNE positivity. Empty image ranks omit their subtracted blocks.

If R_b^{CIT} is the particular earlier CIT25 lift, there is a unique invertible C_b on quotient values and some A_b with $R_b^{\text{CIT}} = R_b C_b + Y_{\text{inv}} A_b$. The exact minimum gives

$$Q_b^{\text{CIT}} = C_b^* Q_b^E C_b, \quad \det Q_b^{\text{CIT}} = |\det C_b|^2 \det Q_b^E. \quad (\text{RQB15})$$

The coefficients are obtained by solving the full frame Y_W . They are fixed as D or the source-order pair changes. Thus this new calculation transports to the original CIT25 factor with its actual squared frame determinant; it does not silently choose a different quotient volume.

For complete arithmetic rows at target degree j keep every original moment $\mu_d = \sum_{u,w=0}^8 a_{u,w} \beta_{8,u,w}^d$, and put

$$\begin{aligned} \beta_{j,a,b} &= j(1/2 - \delta - i\gamma) + 2a\delta + 2bi\gamma, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{-1,t} &= 0, \quad b_{0,t} = \mathbf{1}_{t=0}, \quad b_{n+1,t} = i^{-1}(b_{n,t-1} - c'_j b_{n,t}) - n(n-1 + s_1/2) b_{n-1,t}, \\ t_n &= \left(\frac{\sum_t b_{n,t} f_{a,t}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_t b_{n,t} \beta_{j-8,u,w}^t}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,w}. \end{aligned} \quad (\text{RQB16})$$

The first recursion follows by differentiating $E_A F_a = N_a$ $n+v$ times and isolating the term with μ_v . The second is the original monic Gamma recurrence after $y = (S' - c'_j)/i$, with $1/i = -i$ retained. Thus t_n, v_n are the complex-linear evaluations in the original orthonormal frame. The measure is exactly

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad s = s_1 \in \{1, j\}.$$

No α_A is inserted in the division rows: it belongs to the KRC coefficient transport already specified in RQB2.

Stack these rows through D as T_D, V_D , and put $\mathcal{T}_D = \text{diag}(T_D, 4)$, $\mathcal{V}_D = \text{diag}(V_D, 4)$. Define the literal full raw frames

$$Z_{\text{inv},D} = [\mathcal{V}_D, \mathcal{T}_D Y_{\text{inv}}], \quad Z_{W,D} = [Z_{\text{inv},D}, \mathcal{T}_D R_b], \quad G_{X,D} = Z_{X,D}^* Z_{X,D}. \quad (\text{RQB17})$$

Every lower evaluation occurs. Both frames are independent: a zero raw value modulo the lower evaluations gives a zero coefficient class by the original upper Vandermonde at $D + v \geq q_j - 1$, and then the coefficient frames in RQB13 are independent. Full elimination, with the complete old lower and invariant coefficients, therefore proves

$$Q_b^E(D) = (\mathcal{T}_D R_b)^* (I - P_{Z_{\text{inv},D}}) \mathcal{T}_D R_b, \quad \det Q_b^E(D) = \frac{\det G_{W,D}}{\det G_{\text{inv},D}}, \quad P_Z = Z(Z^* Z)^{-1} Z^*. \quad (\text{RQB18})$$

This is the full raw Gamma realization of the original residual factor. It has no independent row-by-row gauge minimization.

6.4 The exact original source denominator and compatibility transport

Let Z_Y be a fixed full basis of Y and retain the complement E_b from RQB13. On the source put $G_0 = D_{k,D}^{(s_0)}$. Its quotient form in these fixed residual lifts is

$$Q_0(D) = E_b^* G_0 E_b - E_b^* G_0 Z_Y (Z_Y^* G_0 Z_Y)^{-1} Z_Y^* G_0 E_b. \quad (\text{RQB19})$$

Choose the first independent columns Z_Λ of $B_0 Z_{\mathcal{L}}$ and write $C_0 = [Z_\Lambda, B_0 E_b]$, an invertible full frame of $\mathbb{C}^{t_{K,k}}$. The exact same fibre in RQB4 gives

$$\det Q_0(D) = \frac{|\det C_0|^2 \det H_{B,k,D}}{\det(Z_\Lambda^* H_{B,k,D} Z_\Lambda)}. \quad (\text{RQB20})$$

Indeed the source quotient first minimizes over J to obtain the original $H_{B,k,D}$ and then over $B_0 \mathcal{L}$. The adapted full Gram has determinant $|\det C_0|^2 \det H_{B,k,D}$; eliminating its first block proves RQB20. In full matrix form Q_0 is the Schur complement in the lifts $B_0 E_b$. This preserves both the source low overlap $t_{0,k}$ and its further hidden kernel of rank $\ell - \epsilon$.

Let $A_D = B_{\text{cof}}^* \mathcal{D}_D B_{\text{cof}}$ and let C_D be the full correlated pullback from NGB27 on V . The notation A_D/Y , C_D/Y means the attained quotient in the fixed lifts E_b . By RQB3 the further target minimum satisfies

$$0 \prec Q_b^E \preceq A_D/Y \preceq C_D/Y, \quad K_-^{-2} Q_0 \preceq C_D/Y \preceq K_+^2 Q_0. \quad (\text{RQB21})$$

To prove the first comparison, all representatives in $B_{\text{cof}} E_b b + B_{\text{cof}} Y$ are allowed in $B_{\text{cof}} E_b b + W_{\text{inv}}$, while $W_{\text{inv}} \cap \text{im } B_{\text{cof}} = B_{\text{cof}} Y$. The native form on the former fibre is A_D/Y . The remaining comparisons are NGB27 minimized on the same source affine fibre; this establishes their exact typed maps.

Define

$$\omega_b(D) = \frac{\det Q_b^E(D)}{\det(A_D/Y)}, \quad e_b^{\text{cor}}(D) = \log \frac{\det(C_D/Y)}{\det Q_0(D)}, \quad \omega_Z(D) = \frac{\det A_D|_Z}{\det C_D|_Z} \quad (Z = V, Y).$$

The restriction determinants in the last ratio use identical fixed frames. The adapted determinant identity cancels those frames and gives

$$\log \frac{\det Q_b^E}{\det Q_0} = e_b^{\text{cor}} + \log \omega_V - \log \omega_Y + \log \omega_b, \quad 0 < \omega_b \leq 1. \quad (\text{RQB22})$$

Apply the complete PCJ13–16 shared-update proof to the actual nested original source images $Y \subset V$; these are their DNE functional classes modulo the original source lower evaluations. Its source difference rank is exactly r_b by RQB4. The full compatibility-plus-source-complement forms are the unchanged PCJ1 matrices; call their nonnegative logarithms χ_Y, χ_V . Consequently

$$\begin{aligned} \log \frac{\det Q_b^E(D)}{\det Q_0(D)} &= -(\chi_Y(D) - \chi_V(D)) + \log \omega_b(D) + \varepsilon_b(D), \\ -2r_b \log \max(1, K_-) - r_b \log c_+ &\leq \varepsilon_b(D) \leq 2r_b \log \max(1, K_+) - r_b \log c_- . \end{aligned} \quad (\text{RQB23})$$

Here $c_- = \min(1, L_-^2)$, $c_+ = \max(1, L_+^2)$ are the actual NKC3-4/13 constants at this same k, D and source pair. For the signs, PCJ16 says $-\log(\omega_V/\omega_Y) = \chi_Y - \chi_V + \nu$ with $r_b \log c_- \leq \nu \leq r_b \log c_+$. The correlated quotient error is in $[-2r_b \log \max(1, K_-), 2r_b \log \max(1, K_+)]$ by NBG28. Subtracting ν proves exactly RQB23. Since $r_b \leq t_{K,k} = O(k)$, the explicit error is $O_{\text{actual}}(k \log q_j) = o(kq_k)$ at the original four target endpoints. The two full logarithmic losses remain. In particular the proved one-sided native bound is $Q_b^E \preceq K_+^2 Q_0$, retaining RQB20's original source denominator; an inverse estimate has not been assumed.

6.5 Four original rows and the complete retained target budget

For a new row degree $n \geq q_j - v$, use degree-major row order and

$$\begin{aligned} z_{\text{inv},n} &= [\text{diag}(v_n, 4), \text{diag}(t_n, 4)Y_{\text{inv}}], \quad z_{W,n} = [z_{\text{inv},n}, \text{diag}(t_n, 4)R_b], \\ L_{X,n} &= z_{X,n}G_{X,n-1}^{-1}z_{X,n}^*, \\ R_n &= \text{diag}(t_n, 4)R_b - z_{\text{inv},n}G_{\text{inv},n-1}^{-1}Z_{\text{inv},n-1}^*\mathcal{T}_{n-1}R_b. \end{aligned} \tag{RQB24}$$

The row permutation from copy-major order is fixed and unitary. Eliminate the old Z_{inv} block in the full raw frame: its Gram becomes $\text{diag}(G_{\text{inv}}, Q_b^E)$ and its new row becomes $[z_{\text{inv}}, R_n]$. Thus

$$\begin{aligned} L_{W,n} &= L_{\text{inv},n} + R_n Q_b^E (n-1)^{-1} R_n^*, \\ Q_b^E(n) &= Q_b^E(n-1) + R_n^* (I_4 + L_{\text{inv},n})^{-1} R_n, \\ \gamma_{b,n} &:= \log \frac{\det Q_b^E(n)}{\det Q_b^E(n-1)} = \log \det(I_4 + L_{W,n}) - \log \det(I_4 + L_{\text{inv},n}) \geq 0. \end{aligned} \tag{RQB25}$$

For the middle equality, minimize the added squared row norm over the old eliminated variables. Completing that square gives the displayed inverse $I_4 + L_{\text{inv}}$, as can also be checked by multiplication. The first identity and the rectangular determinant lemma give the last one. All zero-energy rows and rank-zero quotients are included; in the latter case $\gamma_{b,n} = 0$.

Put $d = q_j - v - 1$ and retain the original endpoint signs. The exact return is

$$\begin{aligned} \mathcal{V}_b &= -\log \det Q_b^E(d) - \log \det Q_b^E(d+1) + \log \det Q_b^E(d+q_j) + \log \det Q_b^E(d+q_j+1) \\ &= \sum_{n=d+1}^{d+q_j+1} w_n \gamma_{b,n} \geq 0, \quad w_{d+1} = w_{d+q_j+1} = 1, \quad w_n = 2 \text{ otherwise.} \end{aligned} \tag{RQB26}$$

Telescoping both low-to-high intervals proves every weight and sign. The same return for Q_b^{CIT} is equal by RQB15: its constant $2 \log |\det C_b|$ cancels only under these four signs.

Complete $[U_R, U_b]$ by any fixed columns U_o to a full frame $C_T = [U_R, U_b, U_o]$ of the fourfold original kernel-dual space $\mathbb{C}^{4t_{K,j}}$. Define $Q_o(D)$ as its full quotient metric modulo $\text{im } \Phi$ in the lifts U_o . The first restriction in this frame is the exact ITL6 Gram $H_{\text{inv}}(D) = U_R^* \mathcal{H}_D U_R$. Sequential attained minima give

$$|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}}(D) \det Q_b^E(D) \det Q_o(D). \tag{RQB27}$$

For proof, first eliminate the U_R block in $C_T^* \mathcal{H}_D C_T$; its next diagonal Schur block is RQB14. Eliminate that block; the remaining minimum allows precisely $\text{im}[U_R, U_b] = \text{im } \Phi$ and is Q_o . Thus every fixed-frame determinant is displayed. The missing rank is $4t_{K,j} - \text{rank } \Phi$, with no image-completeness assumption.

Each restriction and quotient in RQB27 increases as D increases: the original DNE metric increases by retaining additional orthonormal rows, and taking minima on the same affine fibres preserves that inequality. Therefore all three original weighted returns are nonnegative. The same is true row by row, and HJR13,18,21 give

$$\begin{aligned} \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_o &= 4\widehat{F}_j^{(s_1)}, \\ \mathcal{V}_{\text{inv}}^{\text{target}} &= \mathcal{R} \log \det H_{\text{inv}}, \quad \mathcal{V}_o = \mathcal{R} \log \det Q_o, \quad 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\widehat{F}_j^{(s_1)}. \end{aligned} \tag{RQB28}$$

Here $\mathcal{R}$ has exactly the signs and cutoffs of RQB26. Its $\widehat{F}_j$ is the original fixed-column kernel return, since $H_{B,j,D} = \widehat{H}_{K,j,D}^{-1}$. The inverse changes the primal determinant sign exactly; no equality with the earlier

F_K has been imposed. The frame constant in RQB27 disappears only because the four signs sum to zero. The inequality also holds on every single row, with the right side $4g_{K,N}$ for that target row. Thus the full aggregate budget retains zero-energy rows and does not require a hypothesis about concentration.

For an explicit finite cap already proved for these same measures, retain HJR24–25 at base degree j :

$$\begin{aligned} \widehat{F}_j^{(s_1)} + F_{0,j,s_1} &= \mathcal{F}_{\partial,j}^{(s_1)} + \mathfrak{e}_{j,s_1} + \delta_{K,j,s_1} - F_{\Xi(L),j}^{(s_1)} - \delta_{0,j,s_1}, \\ |\delta_{K,j,s_1}| &\leq \mathcal{E}_{j,s_1}^{\text{FEX}}, \quad |\delta_{0,j,s_1}| \leq E_{j,s_1}^{\text{tr}}, \quad -e_{j,s_1}^- \leq \mathfrak{e}_{j,s_1} \leq e_{j,s_1}^+, \quad F_{\Xi(L),j}^{(s_1)} \geq 0, \\ \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4U_{j,s_1}^{\text{jet}}, \quad U_{j,s_1}^{\text{jet}} = \mathcal{F}_{\partial,j}^{(s_1)} + e_{j,s_1}^+ + \mathcal{E}_{j,s_1}^{\text{FEX}} + E_{j,s_1}^{\text{tr}}. \end{aligned} \quad (\text{RQB29})$$

Every symbol here is the unchanged finite original OAR/FEX/PEC quantity in the complete HJR proof. The exact equality is retained before applying bounds. Both F_{0,j,s_1} and the omitted positive restriction are nonnegative by the original native row calculation. Together with RQB28 this proves RQB29. HJR26's already proved outer first variation gives

$$\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\Delta_j q_j \mathfrak{C}_{\partial} + o(kq_k) = O_{\text{actual}}(kq_k), \quad \Delta_j = 16j - 48. \quad (\text{RQB30})$$

The actual period is fixed, and $j q_j / (k q_k) \rightarrow 1$. This is an aggregate upper bound on two specified positive native volumes, with their full missing quotient retained in RQB28. It does not assign an allocation or a leading value to either one.

Finally transport the calculation directly into the accepted CIT25–26. In its original lifts use RQB15 and choose the fixed full frame X_W there, so $[X_{\mathbb{J}}, I_R, R_b^{\text{CIT}}] = X_W C_W$. Then the earlier complete equality remains literally

$$|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E. \quad (\text{RQB31})$$

Substitution into CIT26 replaces its one retained factor $\log \det Q_b^{\text{CIT}}$ by $\log \det Q_b^E + 2 \log |\det C_b|$. The raw calculation is RQB16–18, its exact original source quotient is RQB19–20, and its two independent native losses are RQB23. In the signed four-return the frame constants cancel and the residual contribution obeys the directed interval

$$-4\widehat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} \leq -\mathcal{V}_b \leq 0. \quad (\text{RQB32})$$

Indeed RQB28 says $\mathcal{V}_b \leq 4\widehat{F}_j - \mathcal{V}_{\text{inv}}^{\text{target}}$. This is precisely the sign of the residual factor in CIT26. Its kernel and source rank are computed by RQB4–12, and all lower-gauge, source-complement, invariant-overlap and fixed-frame terms remain in the exact transport. The common long connecting loss is not replaced by this bound.

6.6 The directed original kernel cap and its source correction

There is a sharper finite version of RQB29 which uses the full original positive observation return before bounding it. All quantities in this paragraph have base degree j , source $s = s_1 \in \{1, j\}$ and $N_0, N_1, N_2, N_3 = q_j - 1, q_j, 2q_j - 1, 2q_j$. Retain the primal sign vector $\sigma = (1, 1, -1, -1)$ in FEX. With $t = t_{K,j}$ and $L_j = \mathbb{C}[S]_{<v}$, the exact FEX31/34 source correction is

$$\delta_{K,j,s} := \widehat{F}_j^{(s)} - F_K^{(s)} = \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_j}^{(s)} + F_{(K_j+L_j)/K_j}^{(s)}. \quad (\text{RQB33})$$

The endpoint scalar $e_{N,s}$ is the actual logarithmic Jacobian between the full original $(K_j + L_j)/L_j$ source and its conductor image in FEX34. It is not a deleted low direction. For full numerical coefficient expressions

retain

$$\begin{aligned}
M_N^A &= \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{a,b=0}^8 |a_{ab}| e^{|\beta_{s,a,b-4}|(H_N+2)} \right\}, \quad m_N^A = \log M_N^A, \\
w_{d,s} &= d!/(s/2)_d, \quad \zeta_{N,s} = \sum_{d=v}^N \log(w_{d,s}/w_{d-v,s}), \\
e_{N,s}^- &= \zeta_{N,s} - 2(N+1-v-t)m_N^A, \quad e_{N,s}^+ = 2tm_N^A, \\
D_s^+ &= e_{q_j-1,s}^+ + e_{q_j,s}^+ - e_{2q_j-1,s}^- - e_{2q_j,s}^-, \\
\mathcal{E}_{j,s}^{\text{FEX}} &= v\mathcal{L}_s + 2 \max_i |\zeta_{N_i,s}| + 2(4q_j+1)m_{2q_j}^A.
\end{aligned} \tag{RQB34}$$

Here H_N is the harmonic sum and $\mathcal{L}_s$ is the original AKS two-window constant at degree j . The explicit alternative FEX33 is available with its unchanged target degree and measure. FEX34 proves $e_{N,s}^- \leq e_{N,s} \leq e_{N,s}^+$ using the whole Gamma Jacobian and the actual conductor norm. The original two low quotient metrics give $0 \leq F_{L_j} \leq v\mathcal{L}_s$ and $0 \leq F_{(K_j+L_j)/K_j} \leq (v-t_{0,j})\mathcal{L}_s$. Thus RQB33 proves $\delta_{K,j,s} \leq D_s^+ + (v-t_{0,j})\mathcal{L}_s$. FEX36–37 independently proves $|\delta_{K,j,s}| \leq \mathcal{E}_{j,s}^{\text{FEX}}$. These two bounds retain the original source mass in their proved Gamma norm ratios and keep $t_{0,j}$ before estimating.

Let $\Phi_{j,s}$ be exactly OAR4's full positive sum $\log(1+\eta_{q_j-1}) + 2 \sum_{N=q_j}^{2q_j-2} \log(1+\eta_N) + \log(1+\eta_{2q_j-1})$. All η_N are the original ROQ constrained rows with their complete conductor-observation covariance. If $U_{j,s}^{\text{old}}$ denotes the retained AKS/KAF kernel cap, OAR3/7 and RQB33–34 give the finite number

$$\begin{aligned}
U_{j,s}^{\text{native}} &:= \min \{ U_{j,s}^{\text{jet}}, \min \{ U_{j,s}^{\text{old}}, \mathcal{F}_{\partial,j}^{(s)} + e_{j,s}^+ - \Phi_{j,s} \} \\
&\quad + \min \{ D_s^+ + (v-t_{0,j})\mathcal{L}_s, \mathcal{E}_{j,s}^{\text{FEX}} \} \}, \\
0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4\hat{F}_j^{(s)} \leq 4U_{j,s}^{\text{native}}.
\end{aligned} \tag{RQB35}$$

Indeed the inner first minimum bounds the original F_K from above, using its full residual subtraction. The second minimum bounds the exact difference $\hat{F}_j - F_K$. Their sum bounds $\hat{F}_j$; HJR25 supplies the other cap. This proves the outer minimum without identifying the two different kernel metrics or assuming a new conditioning estimate. Replacing the full Φ by only its first term recovers the earlier OAR6 partial cap, whereas RQB35 retains all computed positive terms.

Finally FEX37 transfers the proved OVR9 limit superior to this exact fixed-column kernel return. Therefore, with the original $\kappa_* = 32 \log(25 \log 3/(4\pi))$ and the same $\mathfrak{C}_{\partial}$ as in RQB30,

$$0 \leq \limsup_{k \rightarrow \infty} \frac{\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}}}{kq_k} \leq 4 \min \{ \kappa_*, 16\mathfrak{C}_{\partial} \}. \tag{RQB36}$$

For proof divide RQB28 by kq_k , insert $jq_j/(kq_k) \rightarrow 1$ and $(\hat{F}_j - F_K)/(jq_j) \rightarrow 0$, then apply OVR9 separately in each unchanged source order. This proves only the displayed upper bound on the sum of two nonnegative exact volumes. Both remain present in the identity RQB28, with its full missing quotient; no individual limit or allocation is presumed.

7 The four original primitive rows and their full Gamma Gram

Fix the actual quartet, its original invertible period matrix $\mathbf{H}$, logarithm branch, and every coefficient of the nonreduced conductor. All constants below are calculated from these unchanged data. Retain the original PDS polynomial coordinates:

$$\begin{aligned}
k &= 4l+1 \geq 9, \quad j = k+4, \quad q_k = (k+1)^2, \quad Q = (k+5)^2, \\
\beta_* &= \frac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \quad \delta, \gamma > 0, \\
E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{s,a,b}z}, \quad \mu_r = E_A^{(r)}(0), \quad 0 \leq v = \text{ord}_0 E_A \leq 80, \\
\mu_v &\neq 0, \quad D = Q - v \geq 116, \quad s \in \{1, k\}, \quad c = k/2 - 4.
\end{aligned} \tag{PFG1}$$

The order s and center c are the source member of the original pairs $(1, 1)$ and (k, j) . In particular this calculation uses the source center at degree k , even though χ_j occurs in its rows.

Define the literal original polynomials, symbols and translations by

$$\begin{aligned}
L_i(Z, W) &= (H(ZW, Z, W, 1)^T)_i, & b_{i,a,b} &= [Z^a W^b] L_i(Z, W)^4, \\
B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b} z}, & m_{r,i} &= B_i^{(r)}(0) = \sum_{a,b=0}^4 b_{i,a,b} \beta_{4,a,b}^r, \\
\chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & (\mathcal{T}_A p)(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\
(\mathcal{T}_{B_i} p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\
f &= \mathcal{T}_A \chi_j, & \lambda &= \mu_v \left(\frac{Q}{v} \right), \quad \deg f = D, \quad \text{lc } f = \lambda, \quad u_i = \mathcal{T}_{B_i} f.
\end{aligned} \tag{PFG2}$$

The degree assertion follows directly by applying $\mathcal{T}_A S^a = \sum_{r=0}^a \binom{a}{r} \mu_r S'^{a-r}$ to the monic degree- Q polynomial. Its first surviving moment is exactly μ_v . Some $B_i(0)$ may vanish; none of the following arguments presumes that all four row polynomials have degree D .

7.1 Full native frame and the exact PDS matrix

Keep the full measure, mass and recurrence:

$$\begin{aligned}
M_s &= (2\pi)^{s/2}, & dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\
\|p\|_{s,c}^2 &= \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y), & h_{n,s} &= M_s n! (s/2)_n, \\
p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1 + s/2) p_{n-1,s}, \\
\phi_{n,s,c}(S') &= p_{n,s}((S' - c)/i) / \sqrt{h_{n,s}}.
\end{aligned} \tag{PFG3}$$

These are exactly PDS6 and SGT2-5. Their orthogonality can be checked without a mass change: the recurrence gives

$$\sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1 + t^2)^{-s/4} \exp(y \arctan t), \quad \int e^{ty} dm_s(y) = M_s (\cos t)^{-s/2} \quad (|t| < \pi/2).$$

For the second identity, the two Euler Gamma integrals followed by $r = u + v_0$, $x = \log(u/v_0)$ give the Fourier pair $|\Gamma(s/4 + i\xi)|^2 = \Gamma(s/2) \int e^{i\xi x} (2 \cosh(x/2))^{-s/2} dx$. The integrand is holomorphic for $|\Im x| < \pi$ and decays exponentially along each smaller closed strip. Contour translation therefore bounds its Fourier transform exponentially; Fourier inversion and analytic continuation give the displayed identity, with $y = 2\xi$ and precisely the multiplier in PFG3. Multiplying the two generating functions and using this integral gives $M_s (1 - tz)^{-s/2}$. Differentiation under the integral is justified by its exponential moments near $(t, z) = (0, 0)$. Coefficient comparison proves $\int p_{n,s} p_{m,s} dm_s = \mathbf{1}_{n=m} h_{n,s}$. Thus PFG3 is the original orthonormal frame. Its leading coefficient in S' is $i^{-n} / \sqrt{h_{n,s}}$.

Let $\mathbf{B}_{s,c}$ be the monomial coefficient map through degree D , defined by $S'^a = \sum_n (\mathbf{B}_{s,c})_{n,a} \phi_{n,s,c}$. Its entries vanish for $n > a$ and its diagonal is $i^n \sqrt{h_{n,s}}$. Define

$$\mathcal{C} = (\mathbf{B}_{s,c} \text{coeff}_{S'} u_i)_{i=1}^4, \quad \mathcal{G}_{k,s} = \mathcal{C}^* \mathcal{C} = (\langle u_i, u_t \rangle_{s,c})_{i,t=1}^4. \tag{PFG4}$$

All polynomial coefficients and row order here are the original ones.

There is an exact map to the retained PDS correction. Set $\ell_1 = 8k + 24$, $r_a = \mathcal{T}_A(\chi_k S^a)$ for $0 \leq a \leq \ell_1$, and

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = ([S^a] R_i)_{a,i}, \quad C_1 = (\mathbf{B}_{s,c} \text{coeff}_{S'} r_a)_a, \quad H_1 = C_1^* C_1.$$

Every shift of an upper degree- k root is an upper degree- j root; all roots are distinct. Thus each numerator in R_i is divisible by χ_k , and $\deg R_i \leq Q - q_k = \ell_1$. Translation sums commute, so, column by column,

$$C_1 W = C, \quad \boxed{\mathcal{G}_{k,s} = W^* H_1 W.} \quad (\text{PFG5})$$

This is the original PDS8 frame, with its positive degree-zero wedge signs. No transpose in the complex-linear quotient map is replaced by an adjoint. The stars in PFG4–5 belong only to the native Gram.

7.2 The four actual moment pivots

We first prove that the 25 by 4 coefficient matrix $(b_{i,a,b})_{(a,b),i}$ has rank four, including every actual period. The kernel of the Segre substitution in homogeneous polynomials is generated by $Q_0 = t_1 t_4 - t_2 t_3$. Indeed Q_0 is prime by Gauss's lemma applied to this primitive polynomial linear in t_1 . On $t_4 \neq 0$ substitute $t_1 = t_2 t_3 / t_4$; a polynomial vanishing on the Segre chart has identically zero Laurent remainder. Consequently some power of t_4 times it is divisible by Q_0 . Primality and $Q_0 \nmid t_4$ remove that power. This proves the full ideal assertion and its converse. The homogeneous Segre parameters here can be ordered as $(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$; on $t_4 \neq 0$ they give exactly $(ZW, Z, W, 1)$ after scalar multiplication.

If $\sum_i c_i L_i^4 = 0$, the invertible substitution $x = Ht$ therefore gives

$$F(x) = \sum_i c_i x_i^4 = Q_H(x) R_2(x), \quad Q_H(x) = Q_0(H^{-1}x).$$

Suppose $F \neq 0$. If $c_i = 0$, additivity of degree in x_i in the polynomial integral domain forces Q_H to be independent of x_i . Its quadratic matrix then has a zero row and column, contradicting its rank four. Hence all $c_i \neq 0$. The two quadrics Q_H, R_2 have a nonzero common point: restrict R_2 to the Segre surface. If the resulting biform is zero, every point works. Otherwise choose a projective w for which it is a nonzero homogeneous binary quadratic in z , and take a projective root over $\mathbb{C}$. At that common point $\nabla F = R_2 \nabla Q_H + Q_H \nabla R_2 = 0$. But $\nabla F = (4c_i x_i^3)_i$ has no nonzero zero. This contradiction proves the asserted independence.

Order the original 25 nodes lexicographically in (a, b) . They are distinct: equality of two nodes forces both $2(a - a')\delta = 0$ and $2(b - b')\gamma = 0$. Their Vandermonde determinant is the nonzero product of their ordered pairwise differences. Multiplying the independent coefficient columns by this full matrix proves

$$\text{rank}(m_{r,i})_{0 \leq r \leq 24, 1 \leq i \leq 4} = 4. \quad (\text{PFG6})$$

Let $0 \leq t_1 < t_2 < t_3 < t_4 \leq 24$ be the earliest indices at which this successive row rank becomes 1, 2, 3, 4. Retain the actual fixed matrix

$$\mathcal{M} = (m_{t_a,i})_{a,i=1}^4, \quad \det \mathcal{M} \neq 0, \quad T = \sum_{a=1}^4 t_a. \quad (\text{PFG7})$$

The determinant is a calculated fixed moment minor; its nonvanishing follows from the construction. It is not a new conditioning assumption. Every row m_r with $r < t_a$ is in the span of the previous selected rows.

Write $f(S') = \sum_{b=0}^D f_b S'^b$, $f_D = \lambda$. The exact row of coefficients at degree $D - t$ of the four u_i is

$$c_t = \sum_{r=0}^t f_{D-t+r} \binom{D-t+r}{r} m_r = \lambda \binom{D}{t} m_t + \sum_{r < t} f_{D-t+r} \binom{D-t+r}{r} m_r. \quad (\text{PFG8})$$

This is the binomial theorem in each original translation. At $t = t_a$, the second term is a linear combination of the previous selected moment rows. It follows, without deleting any coefficient of f , that

$$\det(c_{t_a,i})_{a,i=1}^4 = \lambda^4 \det \mathcal{M} \prod_{a=1}^4 \binom{D}{t_a}. \quad (\text{PFG9})$$

In particular lower polynomial coefficients cannot cancel this pivot.

The same reasoning applies in the original Gamma frame. The row of $\mathcal{C}$ at index $n = D - t$ is $\sum_{u=0}^t (\mathbf{B}_{s,c})_{D-t,D-u} c_u$. For $t = t_a$, its contribution from $u < t_a$ lies in the preceding moment span, and its diagonal contribution is $i^{D-t_a} \sqrt{h_{D-t_a,s}} \lambda_{t_a}^{(D)} m_{t_a}$. With the row order $D - t_1, \dots, D - t_4$, this proves the exact original minor

$$\det \mathcal{C}_{\{D-t_1, \dots, D-t_4\}, 1:4} = i^{4D-T} \lambda^4 \det \mathcal{M} \prod_{a=1}^4 \binom{D}{t_a} \sqrt{h_{D-t_a,s}}. \quad (\text{PFG10})$$

The ordering of the selected rows is stated explicitly. The phase uses $S' = c + iy$, not a real-coordinate replacement.

7.3 Full finite Gram bounds

The Cauchy–Binet identity, obtained by expanding the two determinants in $\det(\mathcal{C}^* \mathcal{C})$ and collecting the common row subsets, writes it as the sum of squared moduli of all four-row minors. Retaining the actual nonzero minor PFG10 gives

$$0 < \mathcal{L}_{k,s} := |\mu_v \binom{Q}{v}|^8 |\det \mathcal{M}|^2 \prod_{a=1}^4 \binom{D}{t_a}^2 h_{D-t_a,s} \leq \det \mathcal{G}_{k,s}. \quad (\text{PFG11})$$

This proves positivity with a literal finite original-data lower bound.

For the upper bound retain

$$\begin{aligned} A_{\text{abs}} &= \sum_{a,b=0}^8 |a_{ab}|, \quad B_{i,\text{abs}} = \sum_{r,t=0}^4 |b_{i,r,t}| > 0, \quad R_\beta = \sqrt{(1/2 + \delta)^2 + \gamma^2}, \\ \Lambda_{k,s} &= \max\{1, |c| + 2(D + 1 + s/2)\}, \quad \Theta_{a,b,r,t;u,w} = \beta_{8,a,b} + \beta_{4,r,t} - \beta_{j,u,w}, \\ P_{i,k,s} &= \sum_{a,b=0}^8 \sum_{r,t=0}^4 |a_{ab} b_{i,r,t}| \prod_{u,w=0}^j (\Lambda_{k,s} + |\Theta_{a,b,r,t;u,w}|). \end{aligned} \quad (\text{PFG12})$$

These sums are positive since their coefficient columns are nonzero. The complete original expansion is $u_i(S') = \sum_{a,b,r,t} a_{ab} b_{i,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w})$. Its coefficients above degree D vanish by PFG2, exactly.

In the orthonormal y frame, multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$. Through degree D , both weighted shifts have norm at most $D + 1 + s/2$. Hence multiplication by $S' = c + iy$ on degree at most $D - 1$ has norm at most $\Lambda_{k,s}$. Starting at the constant of squared norm M_s gives $\|p\|_{s,c} \leq \sqrt{M_s} \sum_{n=0}^D |[S'^n]p| \Lambda_{k,s}^n$. The weighted coefficient sum is submultiplicative by its exact coefficient convolution and the triangle inequality. Applied to the full expansion just displayed, it proves $\|u_i\|_{s,c} \leq \sqrt{M_s} P_{i,k,s}$. Using the degree- Q product to bound the coefficient sum before its cancellations is a valid upper bound for the unchanged degree- D polynomial. Finally Gram–Schmidt makes a Gram determinant the product of squared orthogonal remainders, each at most the original squared column norm. Therefore

$$0 < \mathcal{L}_{k,s} \leq \det(W^* H_1 W) \leq \mathcal{U}_{k,s} := M_s^4 \prod_{i=1}^4 P_{i,k,s}^2. \quad (\text{PFG13})$$

Every mass, factorial, moment, translation coefficient and root remains in these finite bounds.

7.4 The derived factorial scale and the primitive correction

Here are explicit two-sided bounds at the next scale. Put

$$C_\beta = 6 + 17R_\beta, \quad c_{\text{low}} = 8 \log |\mu_v| + 2 \log |\det \mathcal{M}|, \quad c_{\text{up}} = 8 \log A_{\text{abs}} + 2 \sum_{i=1}^4 \log B_{i,\text{abs}}. \quad (\text{PFG14})$$

These real constants may have either sign except $C_\beta > 6$. For all original source orders in PFG1, the finite inequalities are

$$\begin{aligned} c_{\text{low}} - 8Q - 2(4v + T)(\log Q + 1) &\leq \log \det(W^* H_1 W) - 4 \log M_s - 8Q \log Q \\ &\leq c_{\text{up}} + 8Q \log C_\beta + 8. \end{aligned} \quad (\text{PFG15})$$

For the upper inequality, the original root formula gives $\max_{a,b} |\beta_{n,a,b}| = nR_\beta$. The separate real and imaginary bounds are attained together at the appropriate corner nodes, so this is the stated exact maximum. It follows that $|\Theta| \leq (k+16)R_\beta$. Also $|c| \leq k/2 + 4$, $s \leq k$ and $D \leq Q$ imply $\Lambda_{k,s} \leq 6(Q+1)$, while $k+16 \leq 17(Q+1)$. Thus $P_{i,k,s} \leq A_{\text{abs}} B_{i,\text{abs}} [C_\beta(Q+1)]^Q$. Take logarithms in PFG13 and use $Q \log(Q+1) \leq Q \log Q + 1$ to obtain the upper side of PFG15.

For the lower inequality let $n_a = D - t_a = Q - v - t_a \geq 92$. Since $s \geq 1$,

$$n!(s/2)_n \geq n!(1/2)_n = \frac{(2n)!}{4^n} \geq (n/\exp(1))^{2n}. \quad (\text{PFG16})$$

The product equality separates even and odd factors of $(2n)!$. The last inequality follows by $\sum_{r=1}^{2n} \log r \geq \int_1^{2n} \log x \, dx = 2n \log(2n) - 2n + 1$. The binomials in PFG11 are positive integers at least one. Consequently its logarithm is at least $c_{\text{low}} + 4 \log M_s + 2 \sum_a n_a \log n_a - 2 \sum_a n_a$. For $1 \leq n_a \leq Q$, integration of the derivative of $x \log x$ gives $Q \log Q - n_a \log n_a \leq (Q - n_a)(\log Q + 1)$. Now $\sum_a (Q - n_a) = 4v + T$ and $\sum_a n_a \leq 4Q$. Substitution proves the lower side of PFG15. The integer v and the actual pivot indices are fixed by the period before taking any limit.

In particular, uniformly in the two original choices $s = 1, k$,

$$\log \det(W^* H_1 W) = 4 \log M_s + 8Q \log Q + O_{\text{actual}}(Q), \quad |\log \det(W^* H_1 W)| = O_{\text{actual}}(Q \log Q) = o(kq_k). \quad (\text{PFG17})$$

Indeed the two explicit sides of PFG15 are $O_{\text{actual}}(Q)$, $\log Q \leq Q$, and $4 \log M_s = 2s \log(2\pi) = O(k)$. Also $Q/q_k \rightarrow 1$ and $\log Q/k \rightarrow 0$ on $k = 4l + 1$. The coefficient 8 follows from four original squared polynomial norms; it was derived from the finite pivot and product bounds.

For clarity, the exact receiving identity preserves the original primitive-coordinate complement. Let Z be any fixed full basis of $\ker W^\top$ and let J be a fixed right inverse of $Z^\top$. Then $\ker Z^\top = \text{im } W$, $[W, J]$ is invertible, and completing the original H_1 square gives

$$\det((Z^\top H_1^{-1} \bar{Z})^{-1}) = \frac{|\det[W, J]|^2 \det H_1}{\det(W^* H_1 W)}. \quad (\text{PFG18})$$

Indeed the full frame Gram has determinant $|\det[W, J]|^2 \det H_1$; its first block is $W^* H_1 W$. The Schur block is exactly the attained quotient minimum with constraint $Z^\top f = a$, since $Z^\top J = I$. Its Gram is the inverse displayed in PFG18. This also proves the complex-linear transpose and conjugate-inverse orientation used in PDS9. PFG15–17 now control this particular four-row denominator in the original PDS and PSR calculation. The full fixed-frame determinant $|\det[W, J]|^2$, and the original connecting-coordinate or complement matrix, remain in PFG18.

8 The original relation determinant and its connecting-image return

Keep every original quartet, period, branch, nonreduced conductor, coefficient and norm in PDS1–9, PSG1–23 and SGT1–33. The symbol is fixed as the degree varies. In particular

$$\begin{aligned} k = 4l + 1 \geq 9, \quad q &= (k+1)^2, \quad Q = (k+5)^2, \quad e \in \{0, 1\}, \quad d_e = 8k + 24 + e, \\ v &= \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad N_e = Q - v - 1 + e, \quad s = 1 \text{ or } k, \\ u_a &= \chi_k(S) S^a, \quad r_a = \mathcal{T}_A u_a \quad (0 \leq a < d_e), \\ H_e &= \text{Gram}_{s, k/2-4}(r_0, \dots, r_{d_e-1}), \quad J_{e,s}^\chi = \text{Gram}_{s, k/2, \text{quot}}([u_a])_{a < d_e}. \end{aligned} \quad (\text{PSR1})$$

The source quotient here is by all of $\mathbb{C}[S]_{<v}$, and the target Gram is exactly PDS7. It remains the Gram of the complete polynomials, not a Gram of their leading monomials. The complete measure in both Grams is

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy.$$

8.1 Finite bounds on the actual translated relation Gram

Retain the exact positive factorial band and the finite PSG errors:

$$\begin{aligned} n_a &= q + a - v, \quad m_a = n_a + (s-1)/2, \quad \lambda_a = \mu_v \binom{q+a}{v}, \\ \mathfrak{F}_{k,e,s} &= \prod_{a < d_e} |\lambda_a|^2 (2\pi)^{s/2} n_a! (s/2)_{n_a}, \quad \mathcal{A}_{k,e,s}^- = B_{k,e,s}^- + 2v \log C_E + E_{k,e,s}^{\text{band}}, \\ \mathcal{A}_{k,e,s}^+ &= B_{k,e,s}^+ + E_{k,e,s}^{\text{band}}, \quad \mathcal{M}_{k,e} = 2d_e q \log(4/\pi). \end{aligned} \tag{PSR2}$$

All three finite expressions $B^\pm, C_E, E^{\text{band}}$ are the literal PSG9, PSG15 and PSG20 expressions. For $v = 0$ use $C_E = 1$. The finite estimates below hold for $k \geq \max(9, v-1, \lceil (\pi/2)(\delta^2 + \gamma^2)^{1/2} \rceil + 1)$; this ensures every stated PSG finite condition. Their limiting conclusions concern the unchanged sequence $k = 4l + 1$.

For any fixed admissible radius $0 < a < 1$ in SGT14, keep the full pole, residue, zero-free-disk and diagonal constants of SGT11–25 and set

$$\begin{aligned} \mathcal{E}_{k,e,s}^-(a) &= \mathcal{A}_{k,e,s}^- + \mathcal{L}_{a,N_e,s,d_e}, \quad \mathcal{E}_{k,e,s}^+(a) = \mathcal{A}_{k,e,s}^+ + \mathcal{U}_{a,N_e,s,d_e}, \\ \varepsilon_{k,e,s}^H &= \log \det H_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e}. \end{aligned} \tag{PSR3}$$

Then the finite inequalities and their limiting consequence are

$$\boxed{-\mathcal{E}_{k,e,s}^-(a) \leq \varepsilon_{k,e,s}^H \leq \mathcal{E}_{k,e,s}^+(a), \quad \varepsilon_{k,e,s}^H = o(kq).} \tag{PSR4}$$

Here is the exact map proving this receiving statement. Express the full u_a in the input Gamma frame and delete precisely the rows below v , giving the SGT30 matrix Y_e . The quotient minimum is $Y_e^* Y_e = J_{e,s}^X$. The actual translated coefficient matrix is $C_e = A_{N_e,s} Y_e$, with all entries of $A_{N_e,s}$ exactly SGT10. Thus

$$\log \frac{\det H_e}{\mathfrak{F}_{k,e,s}} - \mathcal{M}_{k,e} = \left(\log \frac{\det J_{e,s}^X}{\mathfrak{F}_{k,e,s}} - \mathcal{M}_{k,e} \right) + \log \frac{\det(Y_e^* A_{N_e,s}^* A_{N_e,s} Y_e)}{\det(Y_e^* Y_e)}. \tag{PSR5}$$

There is no frame change or unmatched scalar in this identity. PSG21 bounds the first parenthesis by $[-\mathcal{A}^-, \mathcal{A}^+]$; SGT25 bounds the second term by $[-\mathcal{L}, \mathcal{U}]$. Their addition proves both finite sides of PSR4. PSG13, PSG16 and PSG20 give $\mathcal{A}^\pm = o(kq)$. SGT26–31 show that the actual second term is $o(kq)$: for each fixed admissible a its scaled limsup is bounded by a constant times $\log(1/a)$, and admissible a can tend to one only after the degree limit. No claim that the fixed- a bounds themselves are $o(kq)$ is needed. This proves PSR4 in both original orders and at both cutoffs. In particular

$$\frac{1}{kq} \log \frac{\det H_e}{\mathfrak{F}_{k,e,s}} \longrightarrow 16 \log(4/\pi). \tag{PSR6}$$

8.2 Absolute expansion with the full factorial band retained

The exact factorial-band identity is

$$\log \mathfrak{F}_{k,e,s} = d_e \log((2\pi)^{s/2}) - d_e \log \Gamma(s/2) + \sum_{a < d_e} \{2 \log |\lambda_a| + \log(n_a!) + \log \Gamma(m_a + 1/2)\}. \tag{PSR7}$$

For a fully finite bound define

$$\begin{aligned} \mathcal{F}_{k,e,s}^{\text{err}} &= d_e |\log((2\pi)^{s/2})| + d_e |\log \Gamma(s/2)| + 2 \sum_{a < d_e} |\log |\lambda_a|| \\ &+ \sum_{a < d_e} (|n_a - q| + |m_a - q|) \log \max(q, n_a, m_a) \\ &+ \sum_{a < d_e} (3 + \log n_a + \log(2m_a) + \log m_a + \tfrac{1}{2} |\log \pi|). \end{aligned} \quad (\text{PSR8})$$

The elementary integer and half-integer factorial estimates proved after PSG22, and integration of $(x \log x - x)' = \log x$, give

$$|\log \mathfrak{F}_{k,e,s} - 2d_e q(\log q - 1)| \leq \mathcal{F}_{k,e,s}^{\text{err}}. \quad (\text{PSR9})$$

Every n_a, m_a is positive. For $s = 1$ the Gamma term in PSR8 is fixed. For $s = k$, apply $\Gamma(m + 1/2) = \sqrt{\pi}(2m)!/(4^m m!)$ with $m = (k - 1)/2$ and PSG11 to obtain $|\log \Gamma(k/2)| = O(k \log k)$. All degree displacements in PSR8 are $O(k)$, while $d_e = O(k)$ and $|\log |\lambda_a|| = O_{\text{actual}}(\log q)$. Hence $\mathcal{F}^{\text{err}} = O_{\text{actual}}(q \log q) = o(kq)$. Consequently the absolute expansion of the unchanged Gram is

$$\log \det H_e = 2d_e q \{\log(4q/\pi) - 1\} + o(kq). \quad (\text{PSR10})$$

The finite absolute error is bounded below by $-\mathcal{F}^{\text{err}} - \mathcal{E}^-(a)$ and above by $\mathcal{F}^{\text{err}} + \mathcal{E}^+(a)$. PSR7 and PSR4 retain every mass, factorial, first nonzero moment and phase-bearing map behind this expansion.

8.3 Dual source and both primitive coefficient maps

For the original full quotient-source metric PDS7 one has exactly

$$G_e = \overline{H_e^{-1}}, \quad \log \det G_e = -\log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} - \varepsilon_{k,e,s}^H. \quad (\text{PSR11})$$

Indeed positive Hermitian H_e has positive real determinant, so conjugation preserves its determinant and inversion reciprocates it. This also proves the interval for the dual residual: $-\mathcal{E}^+(a) \leq -\varepsilon^H \leq \mathcal{E}^-(a)$.

Now retain CIT's original primitive lift P_e and PDI's exact map $\Theta_e = C_e^\top P_e$. Its image is all of $\mathbb{C}^{d_0}$ at $e = 0$ and $\ker W^\top$ at $e = 1$, where W is the complete rank-four PDS8 coefficient matrix. Put

$$\begin{aligned} \Xi_0 &= 2 \log |\det \Theta_0|, & Z &= \Theta_1, & J_Z &= \overline{Z}^\top (Z^\top \overline{Z})^{-1}, & B_Z &= [W, J_Z], \\ \Xi_1 &= \log \det(W^* H_1 W) - 2 \log |\det B_Z|. \end{aligned} \quad (\text{PSR12})$$

All matrices in this display are the actual matrices for the specified source order. Since Z has independent columns, $Z^\top \overline{Z} = \overline{Z}^* Z$ is invertible. Thus $Z^\top J_Z = I$ and $\ker Z^\top = \text{im } W$; the latter follows from $W^\top Z = 0$ and the exact four-dimensional kernel. Consequently B_Z is square and invertible. The original primitive source Gram at both cutoffs is

$$\begin{aligned} H_{\text{src},e} &= \Theta_e^* G_e \Theta_e, \\ \log \det H_{\text{src},e} &= \Xi_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} - \varepsilon_{k,e,s}^H. \end{aligned} \quad (\text{PSR13})$$

The metric equality is PDI2's equality of the full minimum-norm fibres. At $e = 0$ the determinant identity is immediate from invertibility of Θ_0 . At $e = 1$, block Gram factorization in the full frame $B_Z = [W, J_Z]$ gives

$$\det((Z^\top H_1^{-1} \overline{Z})^{-1}) = \frac{|\det B_Z|^2 \det H_1}{\det(W^* H_1 W)}.$$

The left matrix is the conjugate inverse of $Z^* G_1 Z$, by the attained quotient calculation PDS9. Reciprocating this positive determinant and substituting PSR11 proves PSR13. This supplies the exact four-row correction and its frame determinant; neither is declared to vanish. The finite residual interval in PSR13 is precisely $[-\mathcal{E}^+(a), \mathcal{E}^-(a)]$ after subtracting $\Xi_e - \log \mathfrak{F} - \mathcal{M}$.

8.4 Return to the original connecting-image calculation

Use the two unchanged source-order pairs $(s_0, s_1) = (1, 1)$ or $(k, k+4)$, with $s = s_0$ in PSR1–13. Keep CIT17’s exact $\chi_{X_k, D_e}, \chi_{Y_e, D_e}$, original multiplier and inverse constants $K_{\pm}$, and $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$. Write $r_e = d_e - 4e = 8k + 24 - 3e$ and

$$\begin{aligned} \epsilon_e^{\text{lo}} &= -2r_e \log \max(1, K_-) - r_e \log c_+, & \epsilon_e^{\text{hi}} &= 2r_e \log \max(1, K_+) - r_e \log c_-, \\ \mathcal{Z}_{k,e,s_0,s_1} &= \Xi_e - \log \mathfrak{F}_{k,e,s_0} - \mathcal{M}_{k,e} - \chi_{X_k, D_e} + \chi_{Y_e, D_e}. \end{aligned} \quad (\text{PSR14})$$

The actual connecting-image Gram in its original P_e image frame then satisfies the complete finite estimate

$$\begin{aligned} \epsilon_e^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \mathcal{Z}_{k,e,s_0,s_1} \\ &\leq \epsilon_e^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \\ \log \det H_{\text{conn},e} &= \mathcal{Z}_{k,e,s_0,s_1} + o(kq). \end{aligned} \quad (\text{PSR15})$$

For proof CIT17 is the exact equality $\log \det H_{\text{conn},e} - \log \det H_{\text{src},e} = -\chi_{X_k, D_e} + \chi_{Y_e, D_e} + \epsilon_e$, with $\epsilon_e^{\text{lo}} \leq \epsilon_e \leq \epsilon_e^{\text{hi}}$. Insert PSR13. The remaining residual is exactly $\epsilon_e - \mathcal{E}_{k,e,s_0}^H$, so subtracting the two intervals proves every finite sign in PSR15. CIT17 proves $\epsilon_e = O_{\text{actual}}(k \log q)$, and PSR4 proves the other term is $o(kq)$. At $e = 0$, $\chi_{Y_0, D_0} = 0$ because the source-complement projector is zero; at $e = 1$ the complete four-row term χ_{Y_1, D_1} stays in PSR14.

These identities update PDS7–9, PDI2 and CIT17 at their defining cutoffs through their exact coefficient and quotient maps. At later cutoffs the fixed connecting columns still carry CIT18–26’s full metrics and boundary chain; PSR15 assigns no value to the surviving $\chi_{X_k, D_e} - \chi_{Y_e, D_e}$ or to those later cross ratios. The original factors now evaluated are the entire translated relation Gram and its dual. The remaining connecting and invariant factors are displayed in the same original frames, ready for their actual row calculations.

8.5 The four-row estimate in both original primitive sources

PFG5 identifies its full four-row Gram with $W^* H_1 W$. Use precisely its earliest actual moment indices $t_1 < t_2 < t_3 < t_4 \leq 24$, moment matrix $\mathcal{M}$, $T = \sum t_a$, and original constants $C_\beta, c_{\text{low}}, c_{\text{up}}$ of PFG14. Put

$$\begin{aligned} b_{k,s}^{\text{lo}} &= c_{\text{low}} - 8Q - 2(4v + T)(\log Q + 1), & b_{k,s}^{\text{hi}} &= c_{\text{up}} + 8Q \log C_\beta + 8, \\ m_{k,s}^{(4)} &= 4 \log((2\pi)^{s/2}) + 8Q \log Q, & \varepsilon_{k,s}^{(4)} &= \log \det(W^* H_1 W) - m_{k,s}^{(4)}. \end{aligned} \quad (\text{PSR16})$$

PFG10’s exact nonzero moment pivot and PFG12’s complete original coefficient product prove

$$b_{k,s}^{\text{lo}} \leq \varepsilon_{k,s}^{(4)} \leq b_{k,s}^{\text{hi}}, \quad \varepsilon_{k,s}^{(4)} = O_{\text{actual}}(Q), \quad m_{k,s}^{(4)} = O_{\text{actual}}(Q \log Q) = o(kq). \quad (\text{PSR17})$$

Thus the four-row term retained in PSR12 has now been estimated on its actual polynomial frame. In particular define only the exact coordinate factor, retaining its entire value, by

$$\Gamma_0 = 2 \log |\det \Theta_0|, \quad \Gamma_1 = -2 \log |\det[W, J_Z]|. \quad (\text{PSR18})$$

PSR12 gives $\Xi_e = \Gamma_e + e(m_{k,s}^{(4)} + \varepsilon_{k,s}^{(4)})$. Consequently

$$\begin{aligned} \log \det H_{\text{src},e} &= \Gamma_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} + e m_{k,s}^{(4)} + e \varepsilon_{k,s}^{(4)} - \varepsilon_{k,e,s}^H, \\ e b_{k,s}^{\text{lo}} - \mathcal{E}_{k,e,s}^+(a) &\leq e \varepsilon_{k,s}^{(4)} - \varepsilon_{k,e,s}^H \leq e b_{k,s}^{\text{hi}} + \mathcal{E}_{k,e,s}^-(a). \end{aligned} \quad (\text{PSR19})$$

Every term is evaluated in its original coefficient coordinates. At $e = 0$ both four-row terms are absent, as prescribed by the original primitive space. At $e = 1$ they are the full actual four-row determinant with its finite lower and upper errors. Combining PSR9–10 with PSR17 now gives both original sources the absolute expansion

$$\log \det H_{\text{src},e} = \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} + o(kq). \quad (\text{PSR20})$$

No size for Γ_e is asserted by this formula.

For the original connecting image put

$$\tilde{Z}_{k,e,s_0,s_1} = \Gamma_e - \log \mathfrak{F}_{k,e,s_0} - \mathcal{M}_{k,e} + em_{k,s_0}^{(4)} - \chi_{X_k,D_e} + \chi_{Y_e,D_e}. \quad (\text{PSR21})$$

Direct substitution in CIT17 or PSR15 gives the finite return

$$\begin{aligned} \epsilon_e^{\text{lo}} + eb_{k,s_0}^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \tilde{Z}_{k,e,s_0,s_1} \\ &\leq \epsilon_e^{\text{hi}} + eb_{k,s_0}^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \\ \log \det H_{\text{conn},e} &= \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} - \chi_{X_k,D_e} + \chi_{Y_e,D_e} + o(kq). \end{aligned} \quad (\text{PSR22})$$

This follows by retaining the exact residual $\epsilon_e + e\epsilon_{k,s_0}^{(4)} - \epsilon_{k,e,s_0}^H$. Its three previously proved intervals add with precisely the signs shown. All three residuals are $o(kq)$, and the displayed four-row center also is $o(kq)$ by PSR17. This proves the last line.

8.6 Exact common coordinates for the two original source orders

The original Taylor complement in CIT6 is fixed before it is transported to its Gamma native coordinates. Denote its column matrix by P_e^{Taylor} , and the coefficient matrix of the full polynomials r_a in powers of S' by R_e^{pol} . For the source order s let B_s be the original monomial-to-Gamma matrix through degree N_e . CIT3 and the choice after CIT6 give

$$\begin{aligned} U_s &= \text{diag}(1/r!)_{r=0}^{N_e} B_s^T, \quad P_e^{(s)} = U_s^{-1} P_e^{\text{Taylor}}, \quad C_e^{(s)} = B_s R_e^{\text{pol}}, \\ \Theta_e^{(s)} &= (C_e^{(s)})^T P_e^{(s)} = (R_e^{\text{pol}})^T \text{diag}(r!)_{r=0}^{N_e} P_e^{\text{Taylor}}. \end{aligned} \quad (\text{PSR23})$$

The equality follows by multiplying these three explicit matrices: the full B_s^T and its inverse cancel, including every original mass, leading phase and center coefficient. The remaining expression is independent of s . It follows that Θ_e , Z , J_Z , and Γ_e in PSR12 and PSR18 are literally the same coefficient objects for $s = 1$ and $s = k$. The metric comparison uses the identity on P_e^{Taylor} values; no well-conditioned choice of a new complement has been made.

For completeness the same coefficient identity relates the two connecting-image frames. Let $\tilde{T}_j = U_j T_j$ and $\tilde{V}_j = U_j V_j$ be the Taylor columns of the original analytic germs and lower evaluations. These columns are independent of their Gamma presentation, since each is the Taylor series of the same specified function. CIT3 and CIT8 are equivalently

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Taylor}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}} \quad (\text{PSR24})$$

for the complete lower coefficient matrix E_e^{low} . Existence is precisely CIT6–8. Uniqueness of F_e follows from CNC13: an analytic coefficient column whose jet is a lower evaluation represents zero in the original lower quotient only when the coefficient is zero. The two metric presentations of PSR24 therefore give the same F_e . Thus the following ratios compare both source and target metrics on their same original ordered value spaces.

In these fixed coordinates PSR20 yields the quantitative source comparison

$$\log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} = o(kq). \quad (\text{PSR25})$$

Indeed the identical Γ_e and identical explicit absolute center cancel. A fully finite version follows from PSR19: set

$$\begin{aligned} D_e^{\text{ord}} &= \log \mathfrak{F}_{k,e,1} - \log \mathfrak{F}_{k,e,k} + e(m_{k,k}^{(4)} - m_{k,1}^{(4)}), \\ I_{e,s}^- &= eb_{k,s}^{\text{lo}} - \mathcal{E}_{k,e,s}^+(a), \quad I_{e,s}^+ = eb_{k,s}^{\text{hi}} + \mathcal{E}_{k,e,s}^-(a), \\ I_{e,k}^- - I_{e,1}^+ &\leq \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} - D_e^{\text{ord}} \leq I_{e,k}^+ - I_{e,1}^-. \end{aligned} \quad (\text{PSR26})$$

This subtraction retains all original factorials and mass terms. Finally write $\mathcal{X}_e^{(s_0, s_1)} = \chi_{X_k, D_e}^{(s_0, s_1)} - \chi_{Y_e, D_e}^{(s_0, s_1)}$ for the full original compatibility-plus-complement quantities. Subtracting the two exact CIT17 identities on the common frames proves

$$\log \frac{\det H_{\text{conn}, e}^{(k, k+4)}}{\det H_{\text{conn}, e}^{(1, 1)}} = -\mathcal{X}_e^{(k, k+4)} + \mathcal{X}_e^{(1, 1)} + o(kq). \quad (\text{PSR27})$$

The exact omitted residual in this last display is the source ratio in PSR25 plus $\epsilon_e^{(k, k+4)} - \epsilon_e^{(1, 1)}$; PSR26 and CIT17 give its full finite interval by addition. Thus the leading comparison of these original connecting metrics has been reduced, through explicit maps, to the displayed actual independent gauge loss. The loss itself and the later fixed-column return remain the next native arithmetic calculations.